

Causalidad: Parte I

Resumen en 10 min



Daniel Fernández Guzmán, MD
danferguz@gmail.com

- › **Filiación:** UCSUR, EviSalud
- › **Cursó estudios en:** UNSAAC, UCC, UPCH, IECS, Univ Nebrija.
- › **Ha recibido pagos/financiamiento de:** UNSAAC, UCSUR, UPCH, MINSA (DIRESA Cusco), EviSalud, SOPECARD, IETSI
- › No tengo potenciales conflictos de interés con el tema de esta clase

:D

- › ¿Cómo debería ser realizado el estudio que irrefutablemente pruebe causalidad en un paciente específico? Ojo, no en una población, si no en un paciente!
- › ¿Se pueden estimar los efectos causales individuales que la Medicina Personalizada Promete? ¿Mito o realidad?

Contrafactual y resultados potenciales

› Contrafactual

- “(...) todo acontecimiento o toda situación que no ha sucedido en el universo actualmente observable por la investigación humana, pero que podría haber ocurrido en otros universos posibles.”

› Resultado potencial

- Resultado futuro posible.
- Concepto ligado a contrafactual.
- El mundo observado o factual es un resultado potencial realizado.
- El mundo no observado (contrafactual) es un resultado potencial que no ocurrió (pero que pudo ser).

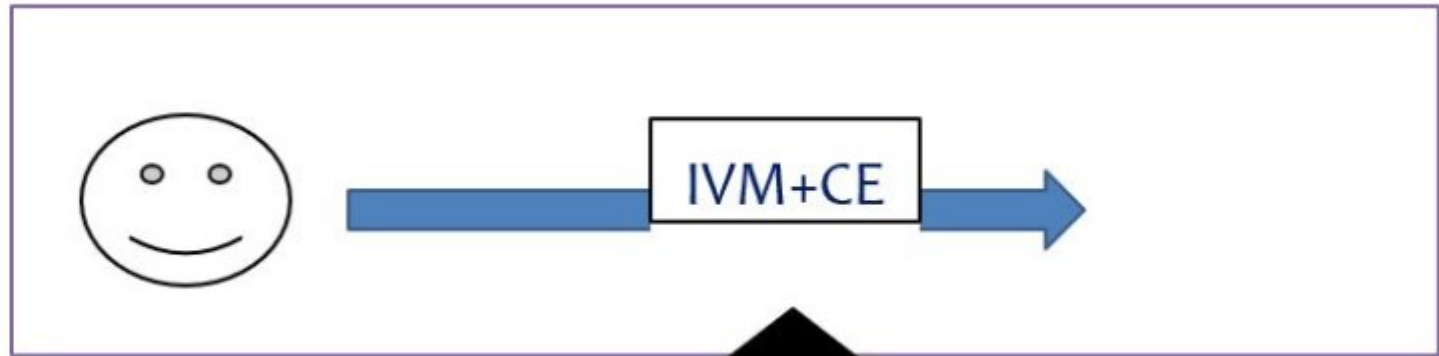
¿Qué necesitamos creer para hacer inferencias causales?

- › Que existen mundos posibles (multiversos?) o, al menos,
- › Que solo existe un mundo (realidad única) pero que hubiera sido un mundo distinto (“mundo posible”) si la causa hubiera sido manipulada.



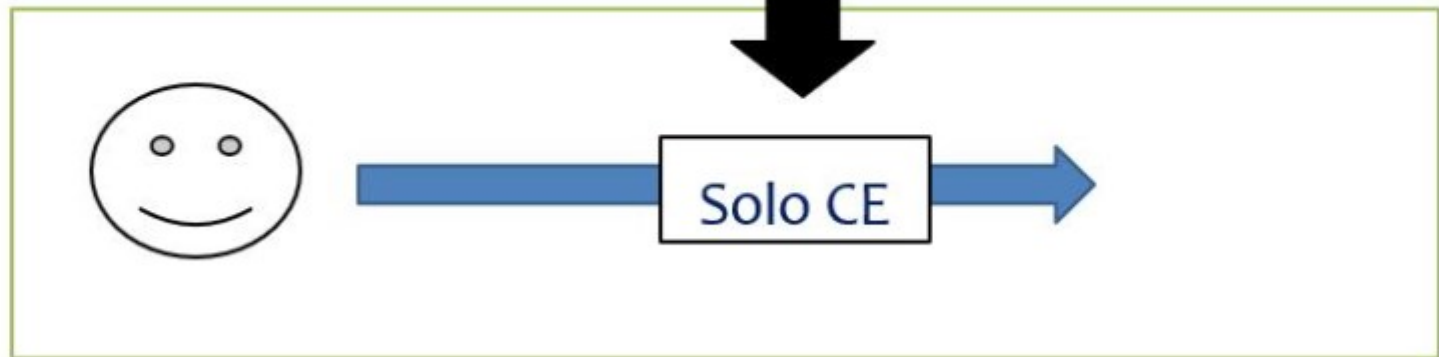
Existen dos multiversos exactamente iguales, salvo por una sola cosa: El tratamiento recibido.

Multiverso 1



(Exactamente iguales salvo en el tratamiento recibido)

Multiverso 2



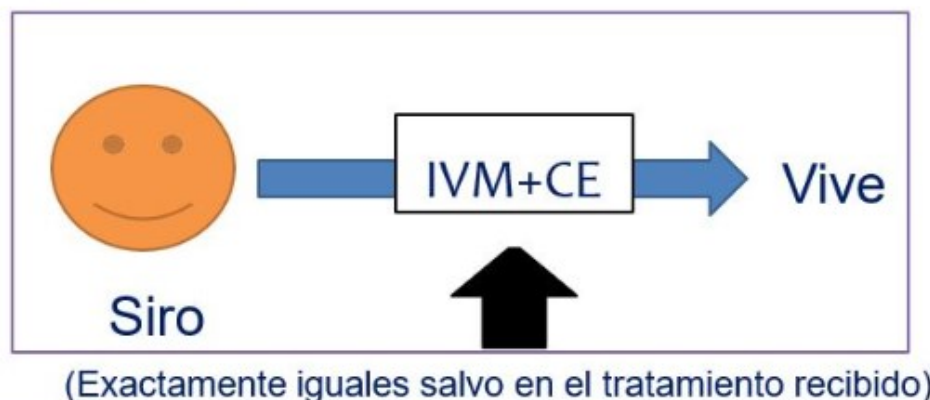
Contrafactuales y resultados potenciales en palabras sencillas...

- *Imaginemos, solo por un rato, que podemos hacer tal experimento y podemos observar y medir lo que pasa en ambos universos:*

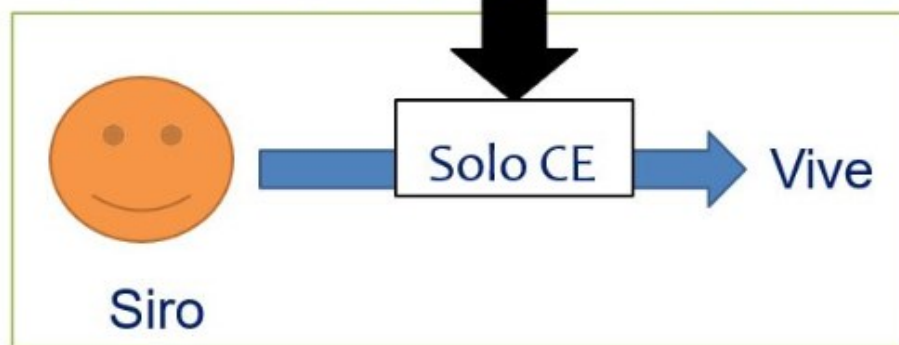


¿Qué pasaría si...?

Multiverso 1



Multiverso 2



Conclusión:

La IVM no tuvo efecto causal en prevenir muerte de Siro.

Hubiera vivido con o sin IVM. Siro estaba destinado a ser 'inmune'.

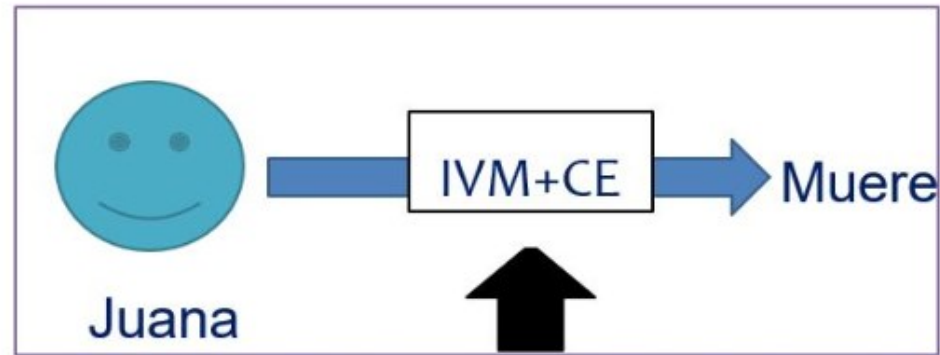
Contrafactuales y resultados potenciales en palabras sencillas...

- *Imaginemos, solo por un rato, que podemos hacer tal experimento y podemos observar y medir lo que pasa en ambos universos:*



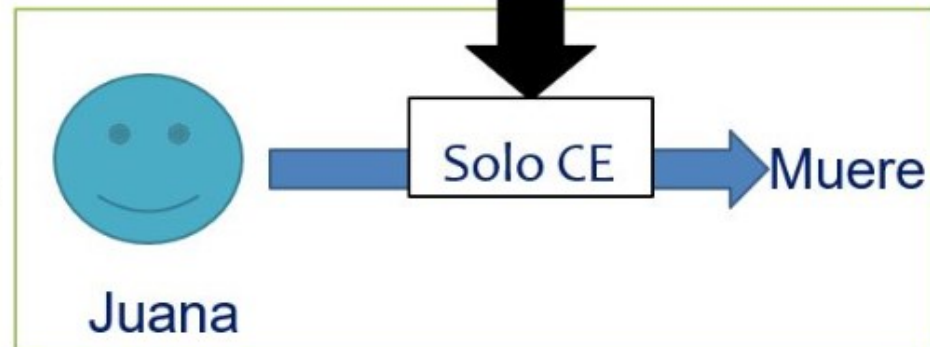
¿Qué pasaría si...?

Multiverso 1



(Exactamente iguales salvo en el tratamiento recibido)

Multiverso 2



Conclusión:

La IVM no tuvo efecto causal en prevenir muerte de Juana.

Hubiera muerto con o sin IVM. Juana estaba destinada a fallecer ('condenada').

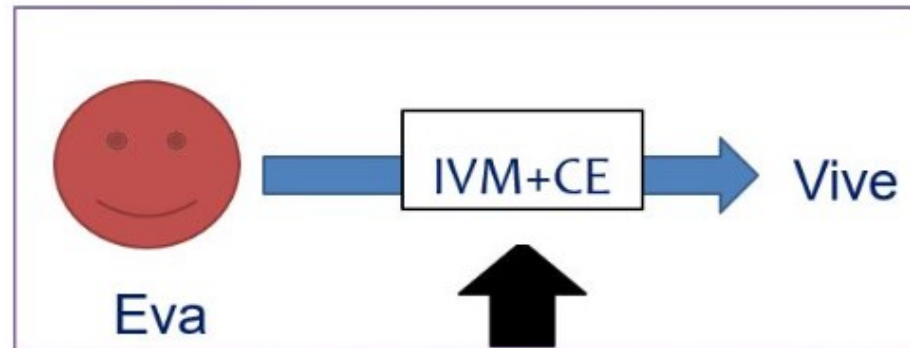
Contrafactuales y resultados potenciales en palabras sencillas...

- *Imaginemos, solo por un rato, que podemos hacer tal experimento y podemos observar y medir lo que pasa en ambos universos:*



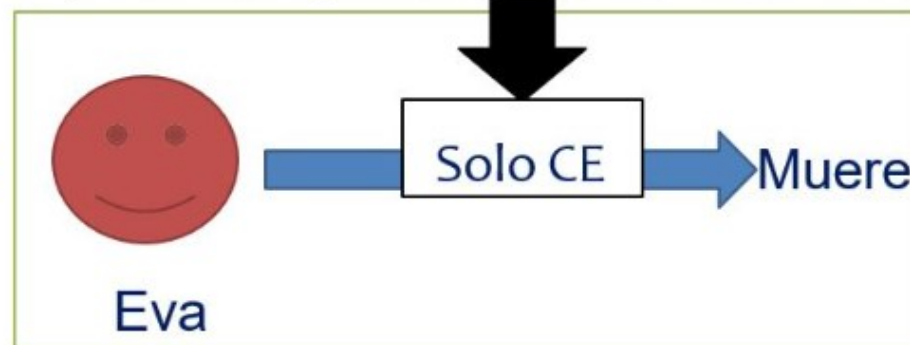
¿Qué pasaría si...?

Multiverso 1



(Exactamente iguales salvo en el tratamiento recibido)

Multiverso 2



Conclusión:

La IVM sí tuvo efecto causal en prevenir la muerte de Eva.
La IVM protegió a Eva de morir.

Contrafactuales y resultados potenciales en palabras sencillas...

- *Imaginemos, solo por un rato, que podemos hacer tal experimento y podemos observar y medir lo que pasa en ambos universos:*



¿Qué pasaría si...?

Multiverso 1



Carlos

(Exactamente iguales salvo en el tratamiento recibido)



Multiverso 2



Carlos



Conclusión:

La IVM sí tuvo efecto causal en ocasionar la muerte de Carlos.
La IVM dañó a Carlos, matándolo.

Notación	Significado
$Y^{x=0} = 0$	Resultado potencial Y cuando <u>configuras</u> la exposición x a no expuesto es igual a 0
$Y^{x=0} = 1$	Resultado potencial Y cuando <u>configuras</u> la exposición x a no expuesto es igual a 1
$Y^{x=1} = 0$	Resultado potencial Y cuando <u>configuras</u> la exposición x a expuesto es igual a 0
$Y^{x=1} = 1$	Resultado potencial Y cuando <u>configuras</u> la exposición x a expuesto es igual a 1



X=0
(IVM + CE)

X=1
(Solo CE)

•Cambiemos de ejemplo

•Resultados Potenciales (Outcome/Desenlace):

- Nivel de calidad de vida (medido por un puntaje psicométrico)

Resultados Potenciales: Efectos individuales

- Efectos causales individuales

Y = Puntaje depresivo X = 1 : Programa Juntos X = 0: No Programa Juntos	Resultado potencial si reciben tratamiento x = 1 $Y_i^{x=1}$	Resultado potencial si reciben tratamiento x = 0 $Y_i^{x=0}$	Efecto individual $Y_i^{x=1} - Y_i^{x=0}$
Siro	10	12	-2
Juana	5	35	-30
Eva	10	46	-36
Carlos	5	40	-35

- Lo que quisiéramos tener...

Individuo	$Y_{i}^{x=1}$	$Y_{i}^{x=0}$
Siro	10	12
Juana	5	35
Eva	10	46
Carlos	5	40

Aunque no podemos estimar efectos causales individuales, sí podemos estimar efectos causales en poblaciones si asumimos ciertos supuestos

Lo que realmente tenemos...



Equivalentemente, en la base de datos

Individuo	$Y_{i}^{x=1}$	$Y_{i}^{x=0}$
Siro	10	.
Juana	.	35
Eva	10	.
Carlos	.	40

	A	B	C
1	Individuo	x	Y
2	Siro	1	10
3	Juana	0	35
4	Eva	1	10
5	Carlos	0	40

Si pudiéramos observar ambos mundos posibles:

Y = Días de estancia hospitalaria	Resultado potencial si reciben tratamiento $x = 1$	Resultado potencial si reciben tratamiento $x = 0$	Efecto individual
Siro	10	12	-2
Juana	5	35	-30
Eva	10	46	-36
Carlos	5	40	-35
	Resultado Potencial Promedio si población hubiera recibido $x = 1$	Resultado Potencial Promedio si población hubiera recibido $x = 0$	Efecto promedio en toda la población:
	7.5	33.25	-25.75

Como solo podemos observar un mundo posible:

Y = Días de estancia hospitalaria	Resultado potencial si reciben tratamiento $x = 1$	Resultado potencial si reciben tratamiento $x = 0$	Efecto individual
Siro	10	.	No identificable
Juana	.	35	No identificable
Eva	10	.	No identificable
Carlos	.	40	No identificable
	Resultado Potencial Promedio <u>estimado</u> si población hubiera recibido $x = 1$	Resultado Potencial Promedio <u>estimado</u> si población hubiera recibido $x = 0$	Efecto promedio <u>estimado</u> en toda la población:
	10	37.5	-27.5

-27.5 → es un estimado del efecto causal poblacional real → -25.75

¡Gracias!

Causalidad: Parte II

Resumen en 10 min



Daniel Fernández Guzmán, MD
danferguz@gmail.com

- › **Filiación:** UCSUR, EviSalud
- › **Cursó estudios en:** UNSAAC, UCC, UPCH, IECS, Univ Nebrija.
- › **Ha recibido pagos/financiamiento de:** UNSAAC, UCSUR, UPCH, MINSA (DIRESA Cusco), EviSalud, SOPECARD, IETSI
- › No tengo potenciales conflictos de interés con el tema de esta clase

Supuestos

π

Consistencia

- Significa que un resultado observado para cada individuo es igual al resultado potencial del tratamiento recibido.
 - El resultado potencial de un individuo tratado es igual al resultado potencial del individuo si hubiera sido tratado.
 - El resultado potencial de un individuo no tratado es igual al resultado potencial del individuo si no hubiera recibido el tratamiento.
- En terminos matemáticos y a nivel individual:

Si $A_i = a$ entonces $Y_i^a = Y_i^A = Y_i$

- En términos poblacionales:

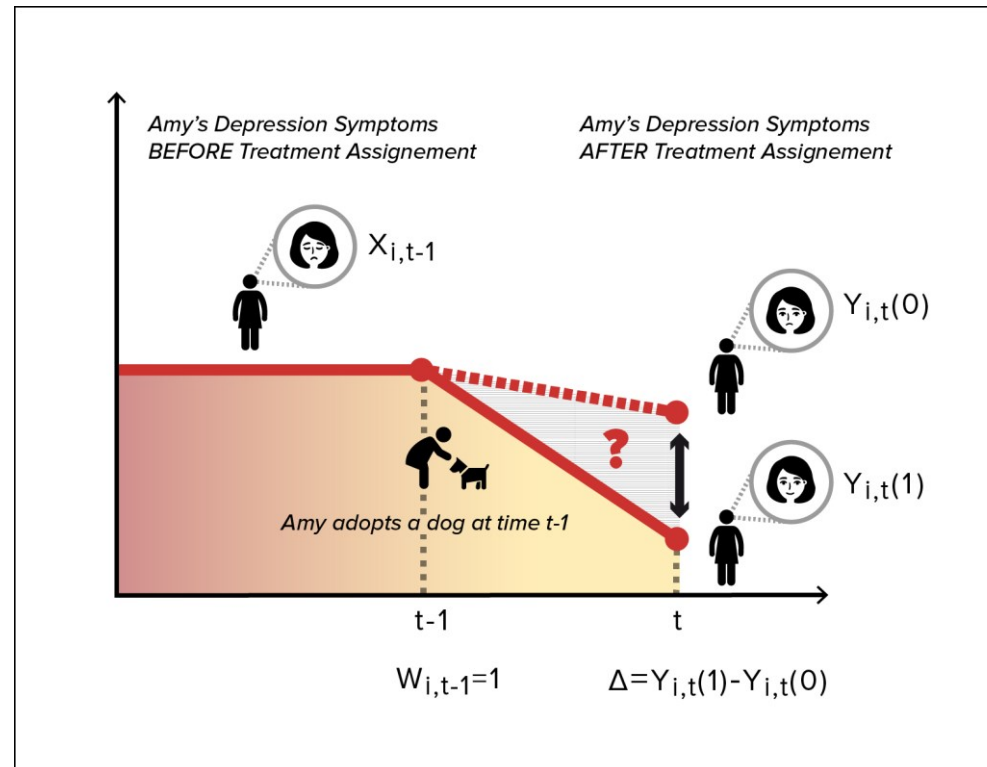
$$Y = Y^A$$

- Otra forma de decirlo:

$$Y = Y^0(1 - A) + Y^1A$$

Consistencia vista de otra manera

- Requiere de la condición: SUTVA (Stable Unit Treatment Value Assumption - Supuesto de Valor de Tratamiento Unitario Estable).
- SUTVA implica dos condiciones:
 - Una sola versión del tratamiento
 - No interferencia entre unidades
- Si este supuesto no se cumple, no es posible **identificar** el efecto causal de **alguna** intervención porque no se tiene intervención bien definida. Tenemos muchas versiones mezcladas de manera difusa!!



Una sola versión del tratamiento

- Existe una intervención bien definida $X = x$.
 - Por ejemplo, dosis exacta, esquema fijo de medicamentos.
 - En otros ejemplos, como el de una cirugía, sería el protocolo exacto de la cirugía.
- Eso implica que cada valor del tratamiento $X = x$ está bien definido.
 - Si hay múltiples versiones del tratamiento $X = x$, el contrafactual Y^x para un individuo i no estará bien definido debido a que el resultado individual depende de la versión del tratamiento x asignado.
 - Por ejemplo, efecto causal del agua en mortalidad. ¿Qué es agua? (cuanto y por qué vía?) con quién se compara?
- Este supuesto matemático es muy ideal y restrictivo, en la práctica se flexibiliza: “Supuesto de irrelevancia de variación de tratamientos”.
 - Por ejemplo, en cirugías, los cirujanos, los lugares donde se realiza la cirugía, suelen ser variables. Para hacer inferencias causales aquí, tenemos que creer que estas “variaciones” son irrelevantes para el efecto del tratamiento.

No interferencia entre unidades

- Cox lo denominó:
 - “no interacción entre unidades” (1958)
- Resultado potencial de un individuo bajo un tratamiento asignado no depende de los tratamientos asignados a otros individuos
- Ejemplos de incumplimiento:
 - **Contagio y vacunas**: Efecto de no haber sido vacunado sobre infección por COVID- 19 cambia dependiendo de la inmunidad de rebaño. Más personas vacunadas cambian la probabilidad que los no vacunados contiguos se infecten.
 - **Spill-over** (“derramamiento”): Programa educativo en intervenidos también influye en resultados potenciales de no intervenidos, ya que ellos aprenden por imitación.

Consistencia parece obvia pero es más problemática de lo que parece

- En variables no manipulables es difícil de justificar consistencia.
- Algunas posturas plantean decisiones más extremas: “No causalidad sin manipulación” (Holland 1986):
 - Postura predominante.
- ¿Deberíamos buscar efectos causales en variables no manipulables? o
- ¿Solo deberíamos identificar relaciones de asociación (en sentido predictivo) que nos sirvan de marcadores de riesgo?

Does obesity shorten life? The importance of well-defined interventions to answer causal questions

M.A. Hernán  & S.I. Taubman

International Journal of Obesity 32, S8–S14 (2008) | [Cite this article](#)

19k Accesses | 228 Citations | 87 Altmetric | [Metrics](#)

[10.1038/ijo.2008.82](https://doi.org/10.1038/ijo.2008.82)



Annals of Epidemiology
Volume 26, Issue 10, October 2016, Pages 674–680



Commentaries on causal identification

Does water kill? A call for less casual causal inferences

Miguel A. Hernán MD, DrPH  

[Show more](#) ▾

<https://doi.org/10.1016/j.annepidem.2016.08.016>

Asignacion aleatoria fisica y experimentos aleatorizados

- Los experimentos aleatorizados usan la asignacion aleatoria fisica como mecanismo de asignacion al tratamiento.
- Como todo estudio en el mundo real (igual que los estudios observacionales), generan datos con valores perdidos de los contrafactuales.
 - Sin embargo, aleatorizacion asegura que los valores perdidos sean completamente aleatorios.
 - Lo que Rubin denomino Missing Completely at Random (MCAR).
- Por tanto, los resultados observados en cada grupo de tratamiento son una muestra aleatoria simple de los resultados potenciales:
 - Podemos usar estas muestras representativas de ambas poblaciones de contrafactuales para estimar consistentemente los efectos causales poblacionales.
- En otras palabras, la asignacion aleatoria fisica vuelve independiente al tratamiento asignado de los resultados potenciales.
 - De tal forma que los individuos pueden intercambiar sus tratamientos y obtener los mismos estimados de efectos causales.
- Veamos una intuicion de esto con R.

Definición de supuesto de intercambiabilidad

Intercambiabilidad o independencia marginal

El tratamiento asignado es independiente de los resultados potenciales de la población:

$$Y^a \perp\!\!\!\perp A, \forall a$$

En el caso tratamiento binario, esto es equivalente a decir que:

$$(Y^1, Y^0) \perp\!\!\!\perp A$$

- En cualquier caso, esto alude a intercambiabilidad de la distribución conjunta, también llamado **intercambiabilidad completa**.

Confusión

Sesgo de selección

Implicancias de la asignacion aleatoria marginal

- En un experimento aleatorizado de manera marginal, la diferencia observada de medias es igual al efecto promedio del tratamiento en la poblacion.
- Ademas, $ATT = ATE$.
- Por lo tanto, la asignacion aleatoria marginal elimina:
 - Sesgo de confusion/seleccion
 - Sesgo por heterogeneidad de efectos

Definicion 7: Diferencia de medias simple en resultados observados

Una diferencia de medias de resultados observados (DMO) puede ser aproximada comparando el desenlace promedio muestral para el grupo tratado ($D = 1$) con un grupo comparador ($D = 0$).

$$DMO = E[Y^1|A = 1] - E[Y^0|A = 0]$$

- DMO no tiene interpretacion causal porque esta comparando individuos diferentes, no las mismas unidades! Entonces, que esta midiendo?

Implicancias de la asignacion aleatoria marginal

- La DMO es el resultado final de la suma de 2 fuentes de sesgo mas el ATE:

$$\begin{aligned}
 DMO = & \underbrace{\text{ATE}}_{\text{Efecto promedio de tratamiento}} + \\
 & \underbrace{E[Y^0|A=1] - E[Y^0|A=0]}_{\text{Sesgo de seleccion/confusion}} + \\
 & \underbrace{(1 - \pi)(ATT - ATC)}_{\text{Sesgo por efectos heterogeneos del tratamiento}}
 \end{aligned}$$

- Donde π es la proporcion de la poblacion que recibe tratamiento en el mundo factual observado.
- Expresion de color rojo no es posible obtener de los datos.

Positividad

- La probabilidad de recibir cada valor del tratamiento es mayor que cero (es positiva).
- En experimentos aleatorizados (marginales o condicionales) esta condicion se cumple por diseno.
 - Por eso, tambien es conocido como **supuesto de tratamiento experimental**.

Positividad en experimentos aleatorizados marginales

$$Pr[A = a] > 0$$

Positividad en experimentos aleatorizados condicionales u estudios observacionales

$$Pr[A = a|L = l] > 0, \forall l$$

¡Gracias!

Causalidad: Parte III

Resumen en 10 min



Daniel Fernández Guzmán, MD
danferguz@gmail.com

- › **Filiación:** UCSUR, EviSalud
- › **Cursó estudios en:** UNSAAC, UCC, UPCH, IECS, Univ Nebrija.
- › **Ha recibido pagos/financiamiento de:** UNSAAC, UCSUR, UPCH, MINSA (DIRESA Cusco), EviSalud, SOPECARD, IETSI
- › No tengo potenciales conflictos de interés con el tema de esta clase

Una motivación para comenzar

Se obtiene información sobre la incidencia acumulada (IA) de cura de 700 pacientes a quienes se les dio acceso a una nueva medicina. Un total de 350 pacientes eligieron tomar la nueva medicina y 350 decidieron no tomarla. Los resultados del estudio se muestran a continuación.

	Nueva medicina (Incidencia de cura)	No tomaron nueva medicina (Incidencia de cura)	Diferencia en las IA de cura (DIA)
Datos SIN controlar por sexo*	273 curados de 350 (78%)	289 curados de 350 (83%)	DIA crudo total = 78% – 83% = -5% ; p = 0.0001
Varón	81 curados de 87 (93%)	234 curados de 270 (87%)	DIA crudo en varones = 93% – 87% = +6%
Mujer	192 curados de 263 (73%)	55 curados de 80 (69%)	DIA crudo en mujeres = 73% – 69% = +4%
Datos controlando sexo*	Modelo de regresión		 DIA ajustado total = +5% ; p = 0.0040

IA = incidencia acumulada, DIA = diferencia de incidencia acumulada

	Modelo crudo Beta; p value	Modelo ajustado Beta; p value
Nueva medicina		
No toma	0.0% (Ref)	0.00% (Ref)
Toma	-5.0%, p=0.0001	+5%; p=0.0040
Sexo		
Varón	—	0.00% (Ref)
Mujer	—	-19.4%; p=0.051
BIC	706.0111	705.043

Para la variable “nueva medicina”, ¿cuál medida de asociación es la correcta?

Una motivación más...

Se obtiene información sobre la incidencia acumulada (IA) de cura de 700 pacientes a quienes se les dio acceso a una nueva medicina. Un total de 350 pacientes eligieron tomar la nueva medicina y 350 decidieron no tomarla. Los resultados del estudio se muestran a continuación.

	Nueva medicina (Incidencia de cura)	No tomaron nueva medicina (Incidencia de cura)	Diferencia en las IA de cura (DIA)
Datos SIN controlar por PA*	273 curados de 350 (78%)	289 curados de 350 (83%)	DIA crudo total = 78% – 83% = -5% ; p = 0.0001
Presión arterial baja	81 curados de 87 (93%)	234 curados de 270 (87%)	DIA crudo en PA baja = 93% – 87% = +6%
Presión arterial alta	192 curados de 263 (73%)	55 curados de 80 (69%)	DIA crudo en PA alta = 73% – 69% = +4%
Datos controlando PA*	Modelo de regresión		 DIA ajustado total = +5% ; p = 0.0040

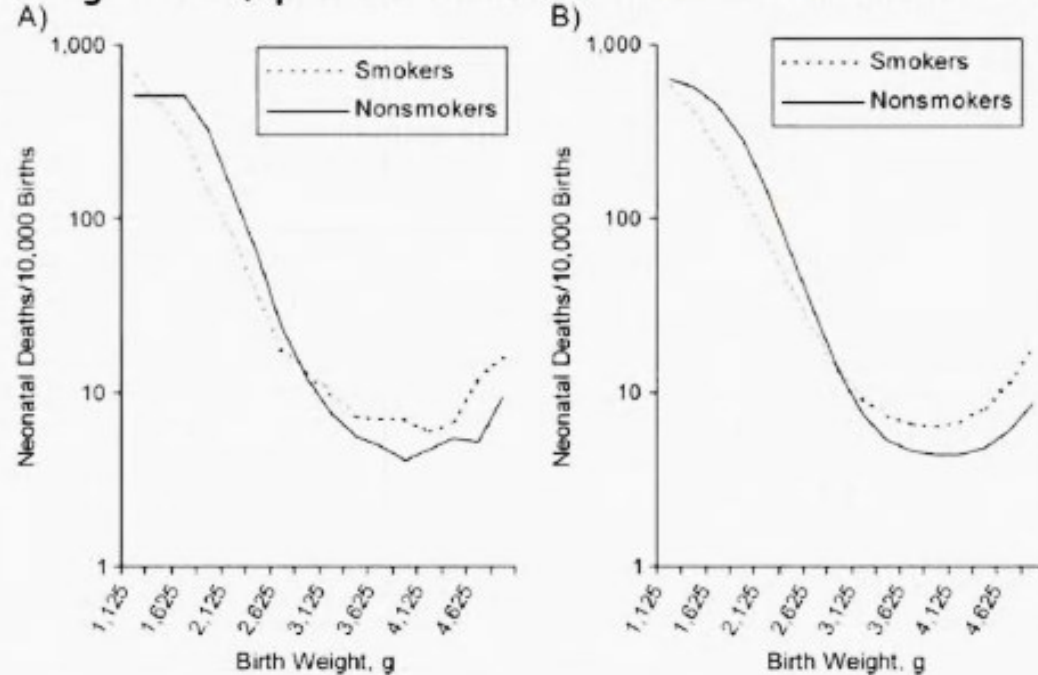
IA = incidencia acumulada, DIA = diferencia de incidencia acumulada

	Modelo crudo Beta; p value	Modelo ajustado Beta; p value
Nueva medicina		
No toma	0.0% (Ref)	0.0% (Ref)
Toma	-5.0%, p=0.0001	+5.0%; p=0.0040
PA		
Baja	—	0.0% (Ref)
Alta	—	-19.4%; p=0.001
BIC	681.011	705.043

Para la variable “nueva medicina”, ¿cuál medida de asociación es la correcta?

Por último, un caso real a modo de motivación: la paradoja del bajo peso al nacer

Curvas específicas para mortalidad infantil al peso de nacer empíricas (A) y calculadas (B) muestran para los fumadores y no fumadores, un ejemplo con los nacimientos entre las semanas 38 y 42 de gestación, que se estima en la fecha de la última menstruación

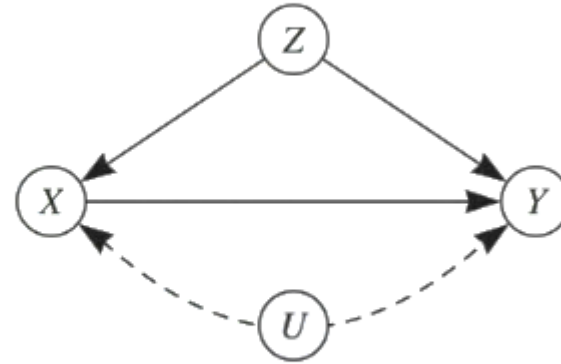


Basso, O. et al. Am. J. Epidemiol. 2009 169:787-797

- Notar que en quienes tienen bajo peso al nacer (<2500 gramos), tener madres fumadoras se asoció consistentemente a menor mortalidad que tener madres no fumadoras.
- Esto también fue consistente entre partos de diferentes semanas de gestación.
- Las tabacaleras dijeron: “**Esto es evidencia de que fumar es benéfico para los bebés con bajo peso al nacer**”
- ¿Qué opinan de esto?

Grafos Acíclicos Dirigidos (DAG) causales

- Son un tipo de diagramas caracterizados por ser:
 - **Dirigidos:** Tienen flechas unidireccionales.
 - **Aciclicos:** El conjunto de flechas no dibuja ciclo (p. ej., una variable no puede causarse a sí misma)
 - **Grafos eulerianos**
- DAG pueden ser usados para múltiples propósitos.
 - Nosotros los usaremos para **causalidad**.
 - Esta clase de DAG es conocido como **Modelo Estructural Causal** porque modelan la **estructura causal** de una pregunta.



Why causal graphs?

- Estimate effect of exposure on disease
- Problem
 - Association measures are biased
- Causal graphs help:
 - Understanding
 - Confounding, mediation, selection bias
 - Analysis
 - Adjust or not
 - Discussion
 - Precise statement of prior assumptions

god-DAG

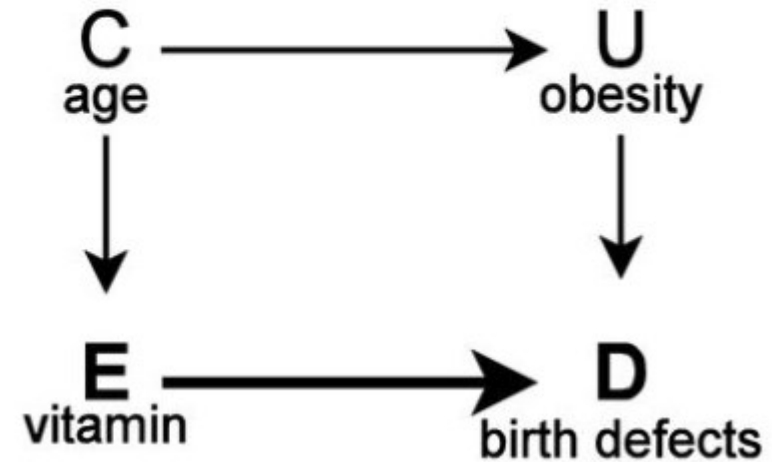
Causal Graph:

Node = variable

Arrow = cause

E=exposure, D=disease

DAG=Directed Acyclic Graph



Read of the DAG:

Causality = arrow

Association = path

Independency = no path

Estimations:

E-D association has two parts:

$E \rightarrow D$ causal effect *keep open*
 $E \leftarrow C \rightarrow U \rightarrow D$ bias *try to close*

Conditioning (Adjusting): $E \leftarrow [C] \rightarrow U \rightarrow D$

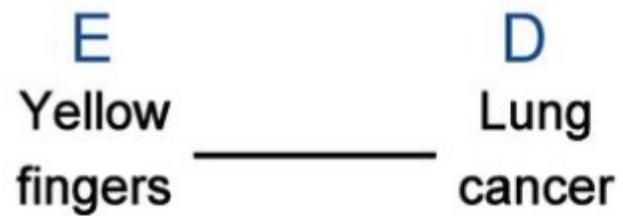
$\rightarrow \rightarrow \rightarrow \rightarrow$ Time $\rightarrow \rightarrow \rightarrow \rightarrow$

¿Qué representa un DAG causal?

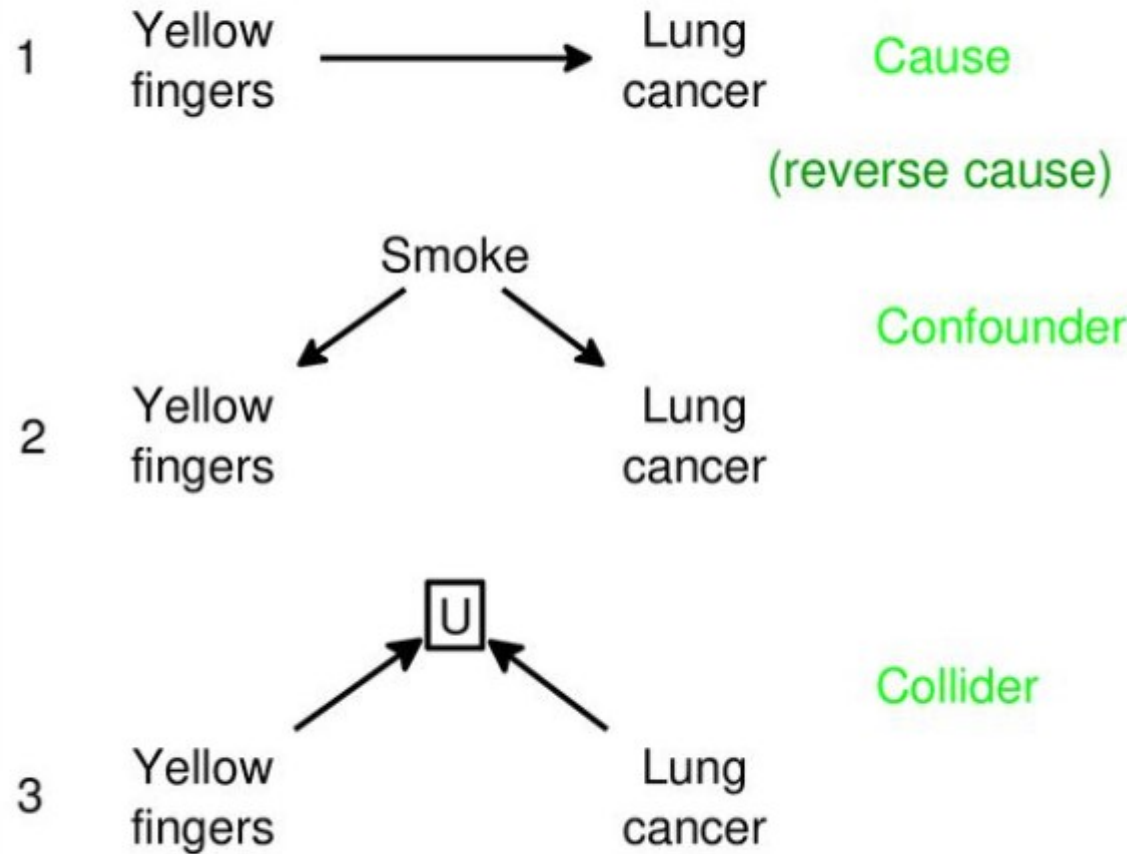
- Los DAG causales reflejan nuestro **conocimiento experto** a priori sobre la estructura causal de un problema.
 - Permiten explicitar transparentemente nuestras **creencias**.
 - Es una representación teórica del estado del arte del conocimiento sobre el fenómeno que estás estudiando.
- Las fuentes de conocimiento pueden tener diferentes grados de credibilidad según el nivel de evidencia que lo sustente.
 - A veces es tan obvio que no requiere publicaciones científicas: Exponer directamente la piel del brazo a una llama de fuego durante 10 minutos causa una quemadura de tercer grado.
 - Otras veces no es tan obvio y requiere evidencias sólidas: La ivermectina causa reducción de mortalidad en pacientes con COVID-19.
- Nuestro conocimiento experto es una mezcla de creencias con diferentes grados de credibilidad y diferentes niveles de evidencia que lo sustenten.
 - Como toda forma de conocimiento es falible.
 - Se actualiza con la evidencia generada en el estudio.

Association and Cause

Association



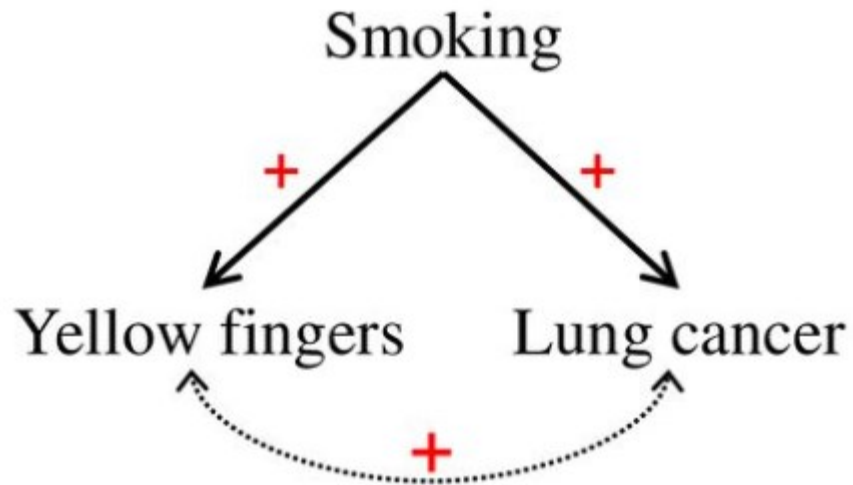
3 possible causal structure



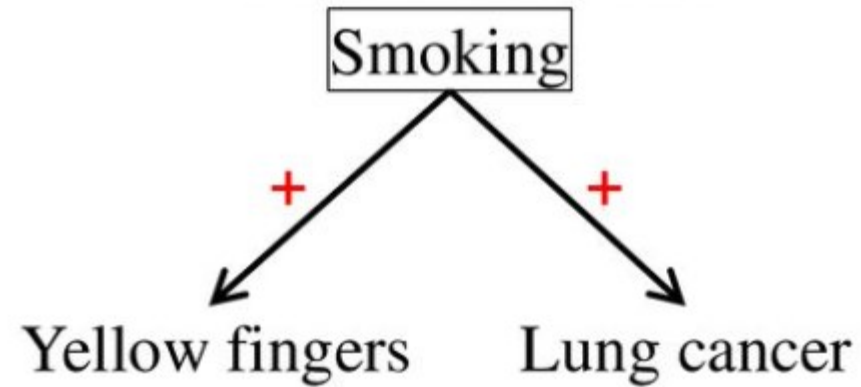
+ more complicated structures

Confounder idea

A common cause



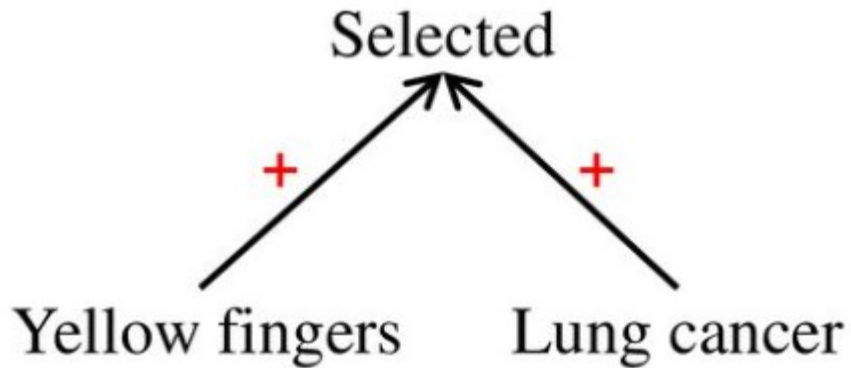
Adjust for smoking



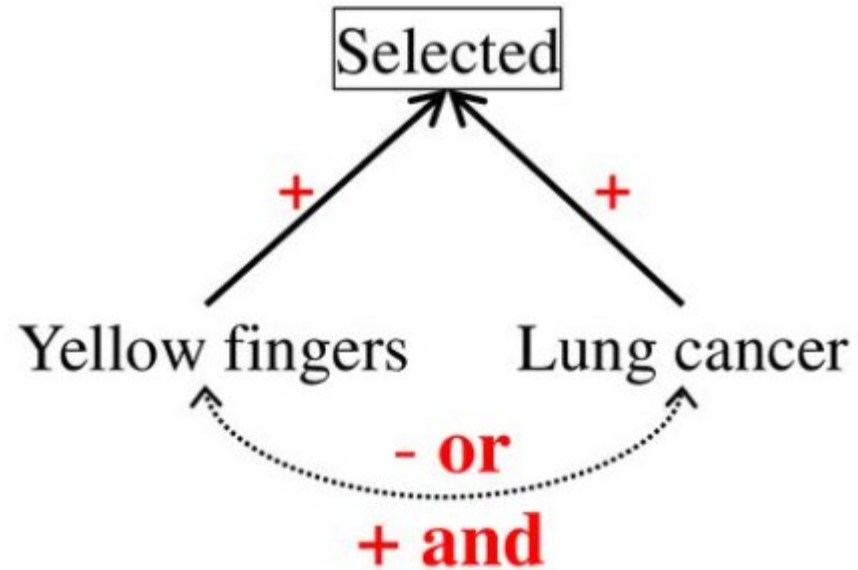
- A confounder induces an association between its effects
- Conditioning on a confounder removes the association
- Condition = (restrict, stratify, adjust)
- Paths

Collider idea

Two causes for selection to study



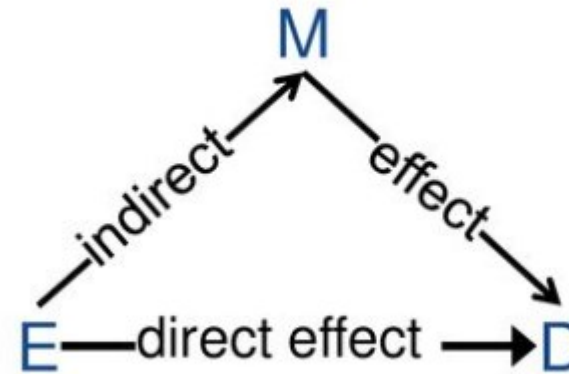
Selected subjects



- Conditioning on a collider induces an association between its causes
- "And" and "or" selection leads to different bias

Mediator

- Have found a cause (E)
- How does it work?
 - Mediator (M)
 - Paths



Total effect = indirect + direct

$$\text{Mediated proportion} = \frac{\text{indirect}}{\text{total}}$$

Use ordinary regression methods if:

no E-M interaction and

collapsible effect measures

Otherwise, need new methods

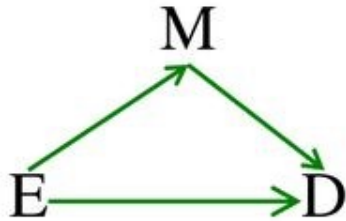
Concepts: Summing up

Associations visible in data. Causal structure from outside the data.

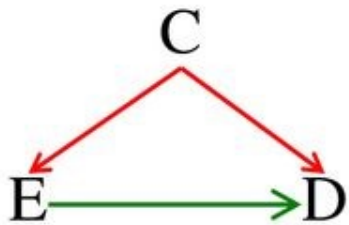
DAG: no arrow means independence



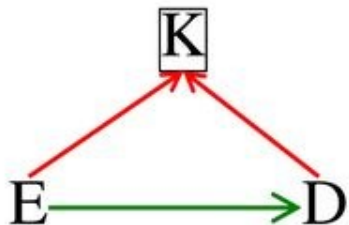
Cause



Cause with Mediator



Cause with Confounder



Cause with Collider

Regression before DAGs

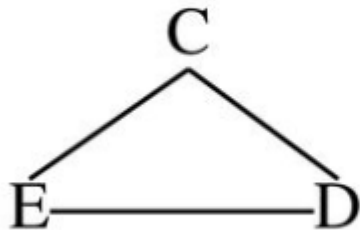
Risk factors for **D**:

Variable	OR	Comments
E	2.0	
C	1.2	Surprisingly low association

Use statistical criteria
for variable selection

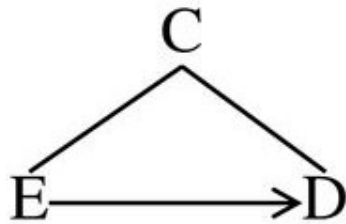
Report all variables in
the model as equals

Association



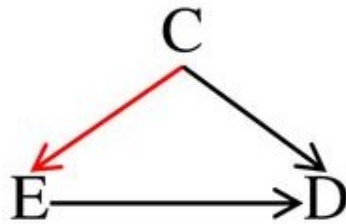
Both can be
misleading!

Statistical criteria for variable selection



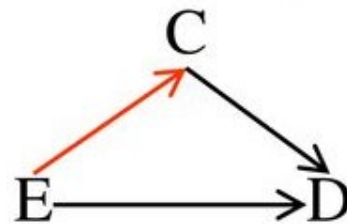
- Want the effect of E on D (E precedes D)
- Observe the two associations C-E and C-D
- Assume statistical criteria dictates adjusting for C (likelihood ratio, Akaike (赤池 弘次) or 10% change in estimate)

The undirected graph above is compatible with three DAGs:



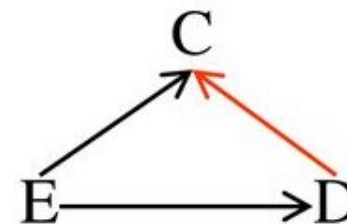
Confounder

1. Adjust



Mediator

2. Direct: adjust
3. Total: not adjust



Collider

4. Not adjust

Conclusion: The data driven method is correct in 2 out of 4 situations
Need information from outside the data to do a proper analysis

DAGs variable selection: close all non-causal paths

Reporting variable as equals:

Association versus causation

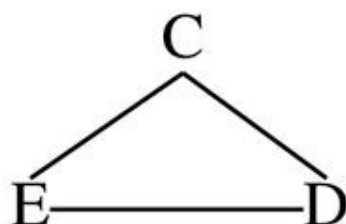
Risk factors for **D**:

Variable	OR	Comments
E	2.0	
C	1.2	Surprisingly low association

~~Use statistical criteria
for variable selection~~

~~Report all variables in
the model as equals~~

Association

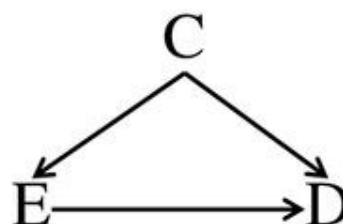


Symmetrical

C is a confounder for E-D

~~E is a confounder for C-D~~

Causation



Directional

C is a confounder for E→D

E is a **mediator** for C→D

Base adjustments
on a DAG

Report only
the E-effect

or use different models
for each variable

Exercise: report variables as equals?

Risk factors for **Fractures**

Variable	OR	Comments (surprises)
Diabetes 2	2.0	
Physical activity	1.2	Protective in other studies?
Obesity	1.0	No effect?
Bone density	0.8	

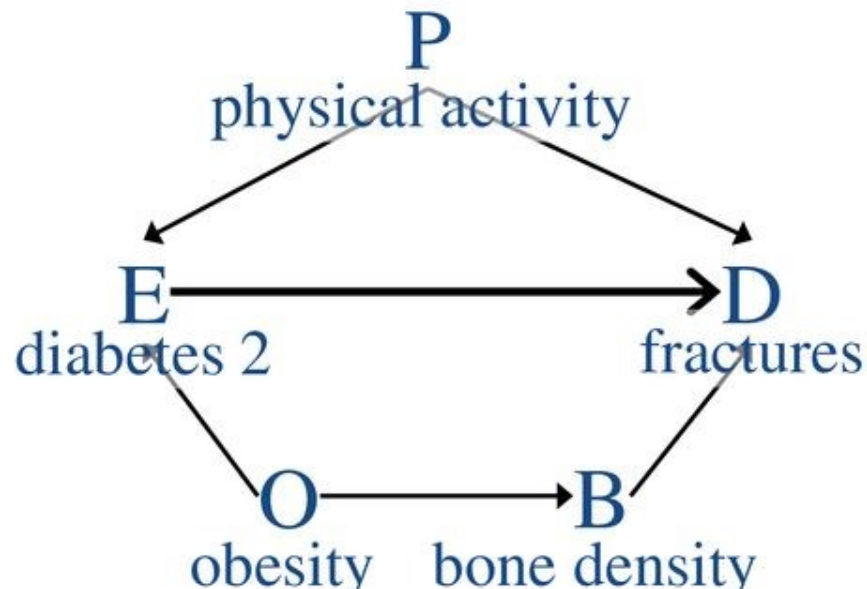
Interpret as effect of:

Diabetes adjusted for all other vars.

Phy. act. adjusted for all other vars.

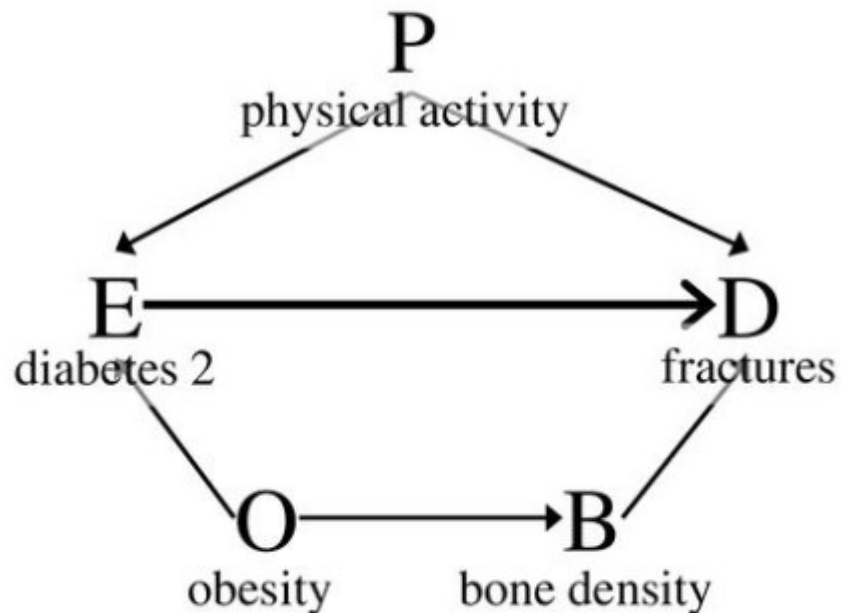
Obesity adjusted for all other vars.

Bone d. adjusted for all other vars.



1. P is a confounder for $E \rightarrow D$, but is E a confounder for $P \rightarrow D$?
2. Which effects are reported correctly in the table?

Variable	OR	Comments (surprises)
Diabetes 2	2.0	
Physical activity	1.2	Protective in other studies?
Obesity	1.0	No effect?
Bone density	0.8	



1. P is a confounder for $E \rightarrow D$, but is E a confounder for $P \rightarrow D$?

No, E is in the causal path $P \rightarrow E \rightarrow D$

2. Which effects are reported correctly in the table?

Only the effects of E and B!

P has indirect effect

$P \rightarrow E \rightarrow D$

O has indirect effects

$O \rightarrow E \rightarrow D$ and $O \rightarrow B \rightarrow D$

DAGs dictate **one exposure** and **one outcome**

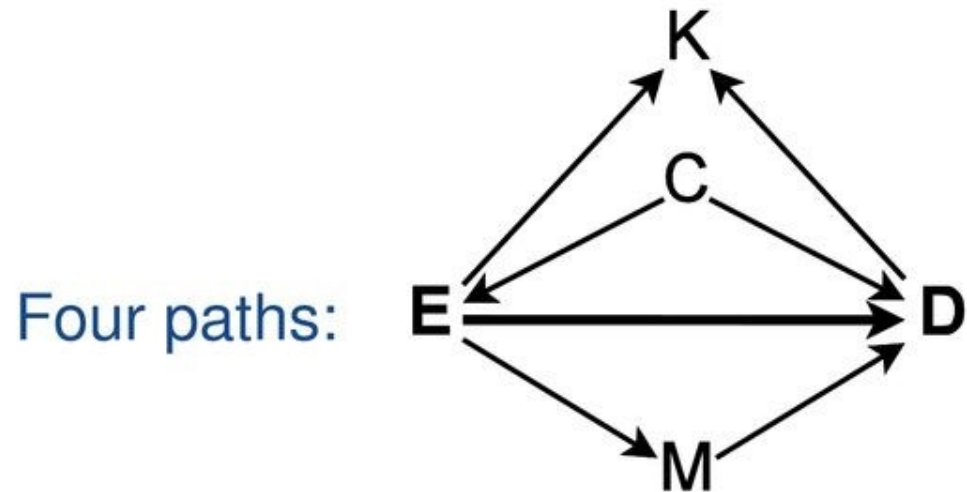
Report adjusted effect of **exposure**

Path definitions

Path: any trail from **E** to **D** (without repeating itself)

Type: causal, non-causal

State: open, closed



	Path
1	$E \rightarrow D$
2	$E \rightarrow M \rightarrow D$
3	$E \leftarrow C \rightarrow D$
4	$E \rightarrow K \leftarrow D$

Goal:

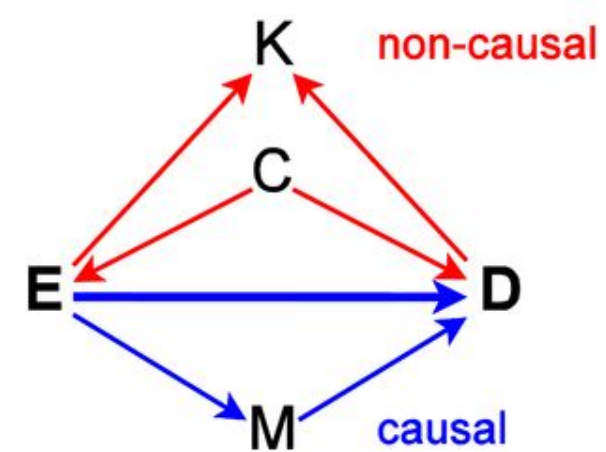
Keep causal paths of interest open

Close all non-causal paths

Four rules

1. Causal path: $E \rightarrow \rightarrow \rightarrow D$

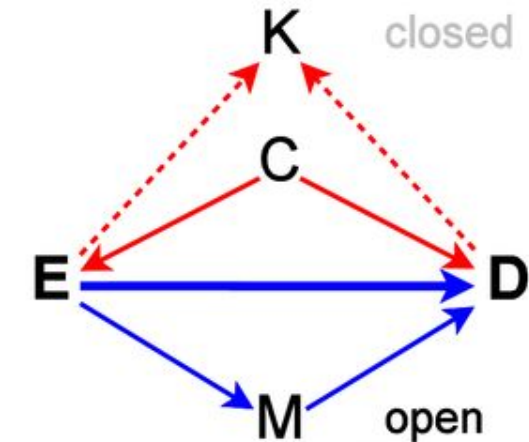
(all arrows in the same direction) **otherwise non-causal**



Before conditioning:

2. Closed path: $\rightarrow K \leftarrow$

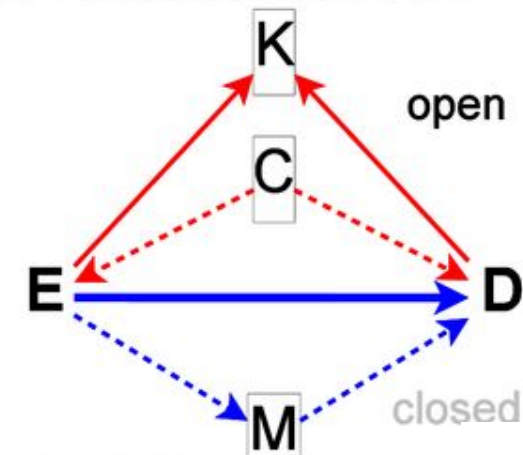
(closed at a collider, **otherwise open**)



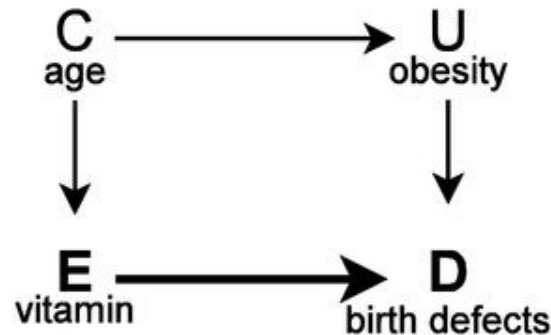
Conditioning on:

3. a non-collider **closes**: $\rightarrow [M] \rightarrow$ or $\leftarrow [C] \rightarrow$

4. a collider **opens**: $\rightarrow [K] \leftarrow$
(or a descendant of a collider)



Vitamin and birth defects



1. Is there a bias in the crude E-D effect?
2. Should we adjust for C?
3. What happens if age also has a direct effect on D?

Unconditional

	Path	Type	Status
1	$E \rightarrow D$	Causal	Open
2	$E \leftarrow C \rightarrow U \rightarrow D$	Non-causal	Open

Bias

This is an example of confounding

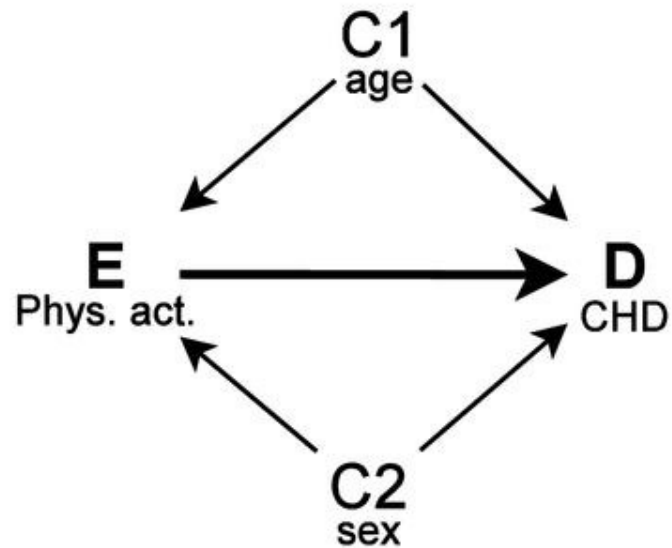
Conditioning on C

	Path	Type	Status
1	$E \rightarrow D$	Causal	Open
2	$E \leftarrow [C] \rightarrow U \rightarrow D$	Non-causal	Closed
3	$E \leftarrow [C] \rightarrow D$	Non-causal	Closed

No bias

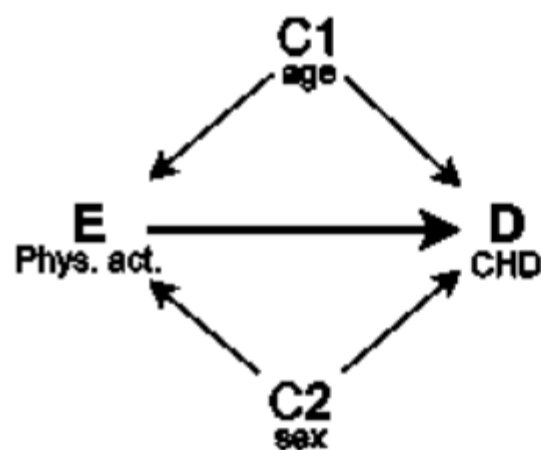
Question:
Is U a confounder?

Exercise: Physical activity and Coronary Heart Disease (CHD)



We want the total effect of Physical Activity on CHD.

1. Write down the paths.
2. Are they causal/non-causal, open/closed?
3. What should we adjust for?



1. We want the total effect of Physical Activity on CHD. What would we adjust for?

Unconditional

	Path	Type	Status
1	$E \rightarrow D$	Causal	Open
2	$E \leftarrow C1 \rightarrow D$	Non-causal	Open
3	$E \leftarrow C2 \rightarrow D$	Non-causal	Open

Bias

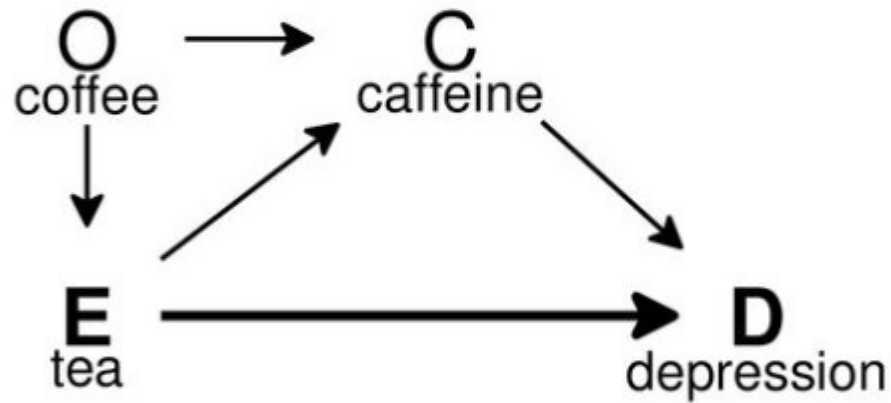
Conditioning on C1 and C2

	Path	Type	Status
1	$E \rightarrow D$	Causal	Open
2	$E \leftarrow [C1] \rightarrow D$	Non-causal	Closed
3	$E \leftarrow [C2] \rightarrow D$	Non-causal	Closed

No bias

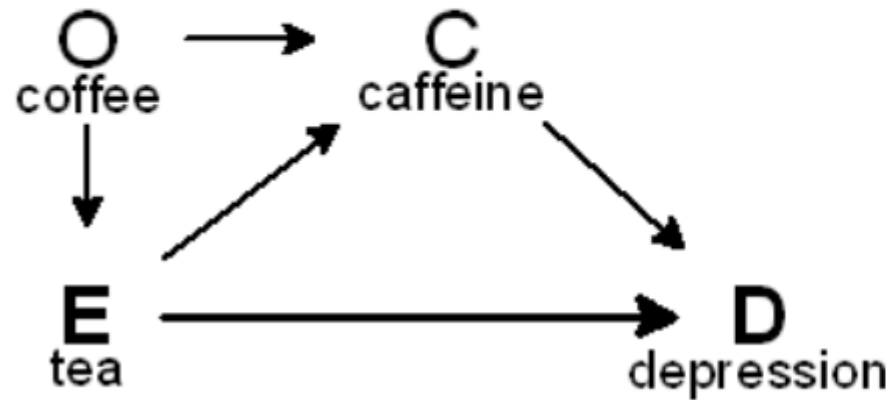
Intermediate variables

Exercise: Tea and depression



1. Write down the paths.
2. You want the **total effect** of tea on depression. What would you adjust for?
3. You want the **direct effect** of tea on depression. What would you adjust for?
4. Is caffeine an intermediate variable or a variable on a confounder path?

Tea and depression



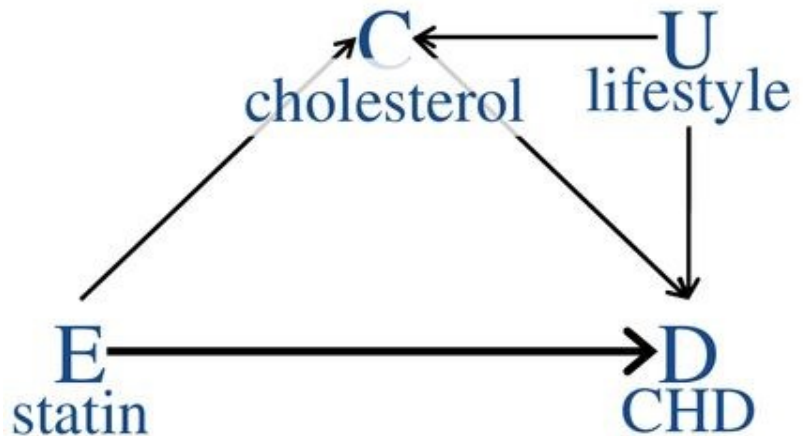
- 1.
2. Total effect: adjust for O
3. Direct effect: adjust for C (and O)
4. Caffeine is **both** intermediate **and** part of a confounder path.

	Path	Type	Status
1	$E \rightarrow D$	Causal	Open
2	$E \rightarrow C \rightarrow D$	Causal	Open
3	$E \leftarrow O \rightarrow C \rightarrow D$	Non-causal	Open

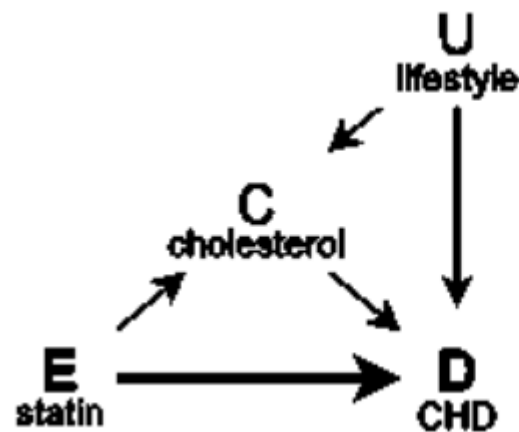
direct
indirect } total



Exercise: Statin and CHD



1. Write down the paths.
2. You want the total effect of statin on CHD. What would you adjust for?
3. If lifestyle is unmeasured, can we estimate the direct effect of statin on CHD (not mediated through cholesterol)?
4. Is cholesterol an intermediate variable or a collider?



1. We want the total effect of statin on CHD. What would you adjust for?
2. Can we estimate the direct effect of statin on CHD (not mediated through cholesterol)?

Unconditional

	Path	Type	Status
1	$E \rightarrow D$	Causal	Open
2	$E \rightarrow C \rightarrow D$	Causal	Open
3	$E \rightarrow C \leftarrow U \rightarrow D$	Non-causal	Closed

No adjustments gives the total effect

Conditioning on C

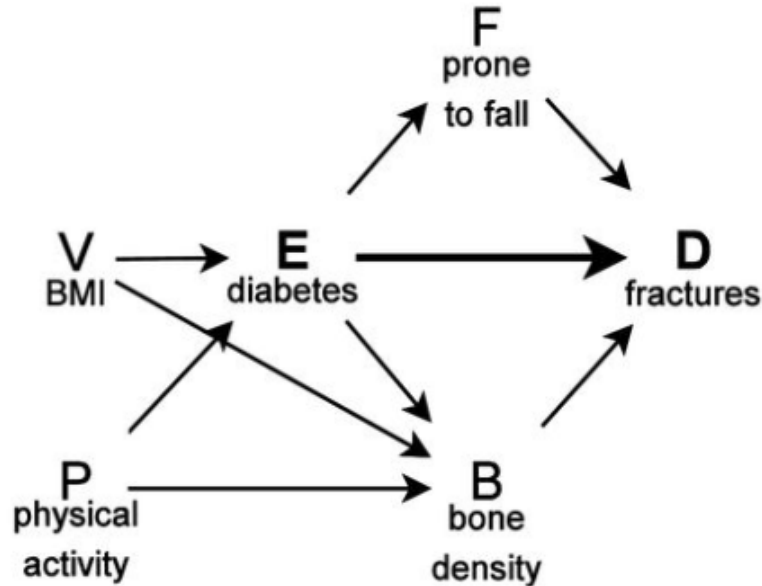
	Path	Type	Status
1	$E \rightarrow D$	Causal	Open
2	$E \rightarrow [C] \rightarrow D$	Causal	Closed
3	$E \rightarrow [C] \leftarrow U \rightarrow D$	Non-causal	Open

Adjusting for C will close path 2 but will open path 3 and give bias!

C is a collider on path 3

Mixed

Diabetes and Fractures



We want the total effect of Diabetes (type 2) on fractures

Conditional

	Path	Type	Status	
1	$E \rightarrow D$	Causal	Open	
2	$E \rightarrow F \rightarrow D$	Causal	Open	
3	$E \rightarrow B \rightarrow D$	Causal	Open	Mediators
4	$E \leftarrow [V] \rightarrow B \rightarrow D$	Non-causal	Closed	
5	$E \leftarrow [P] \rightarrow B \rightarrow D$	Non-causal	Closed	Confounders

Questions:

Paths $\longleftrightarrow$?
 More paths?
 B a collider?
 V and P ind?

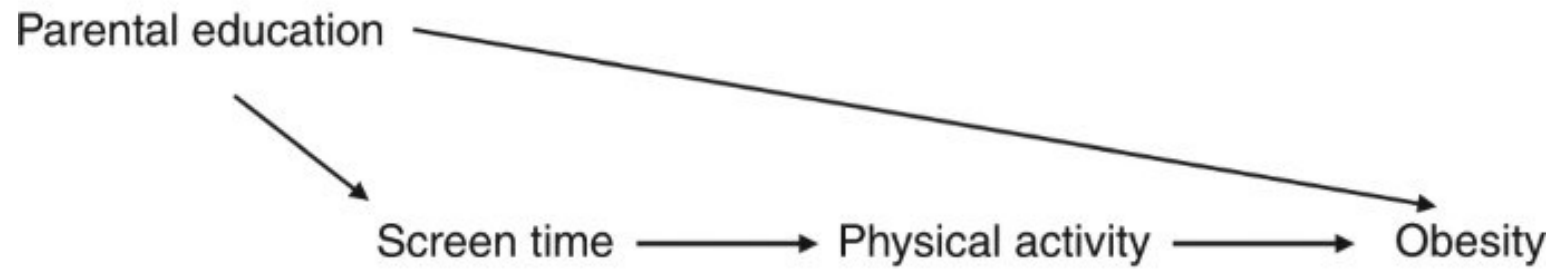
a



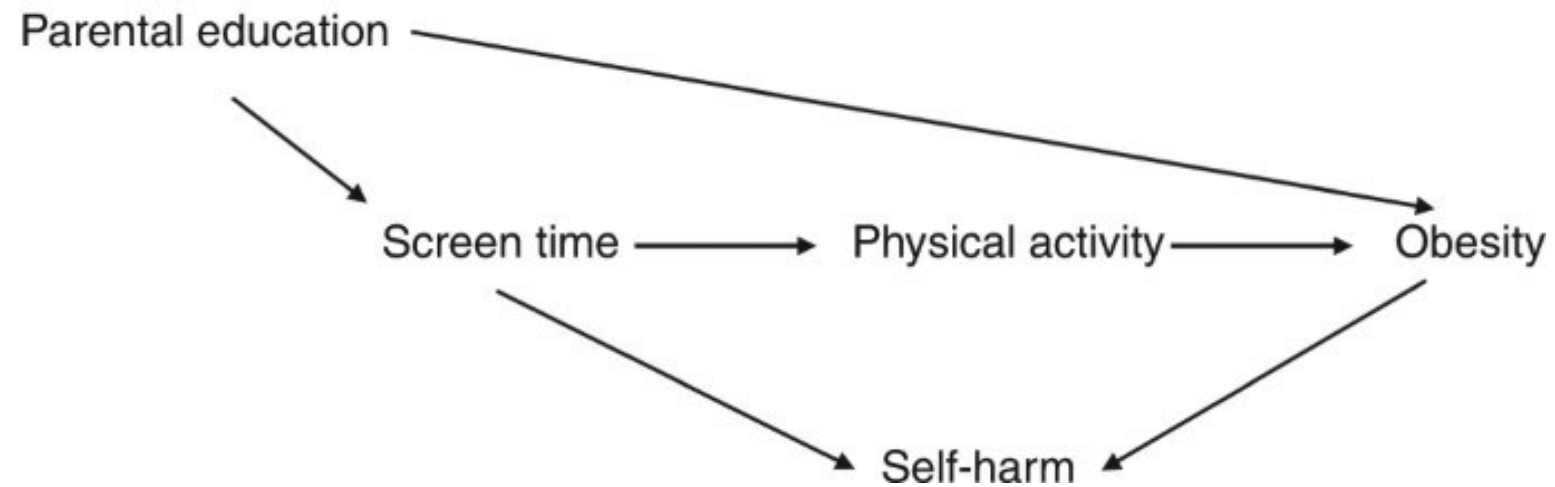
b

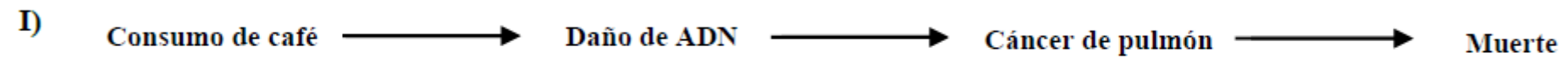


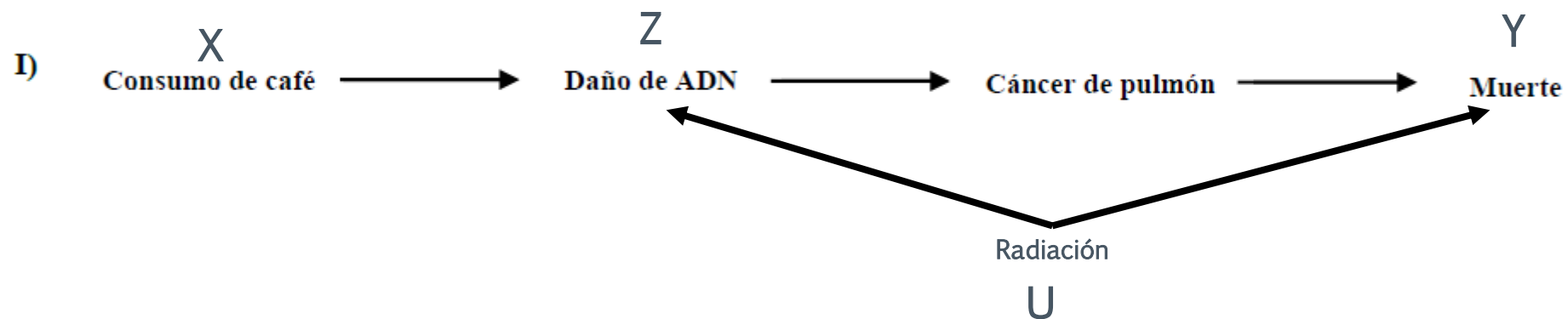
c



d







Paths from X to Y:

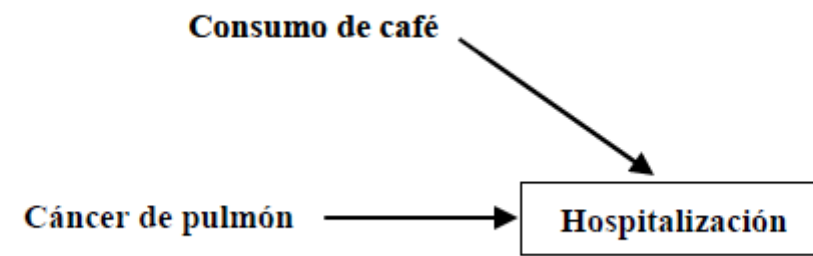
Unconditional

Path	Type	Status	
1 $X \rightarrow Y$	Causal	Open	} total effect
2 $X \rightarrow Z \rightarrow Y$	Causal	Open	
3 $X \rightarrow Z \leftarrow U \rightarrow Y$	Non-causal	Closed	

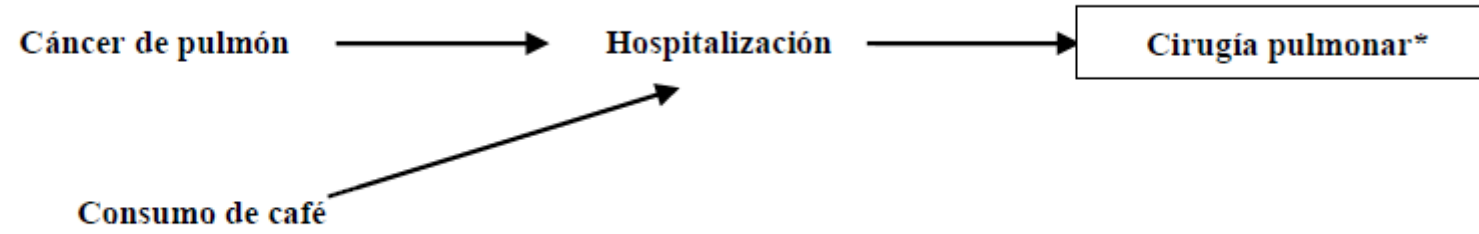
Conditioning on Z

Path	Type	Status	
1 $X \rightarrow Y$	Causal	Open	} direct effect
2 $X \rightarrow [Z] \rightarrow Y$	Causal	Closed	
3 $X \rightarrow [Z] \leftarrow U \rightarrow Y$	Non-causal	Open	} collider bias

II)

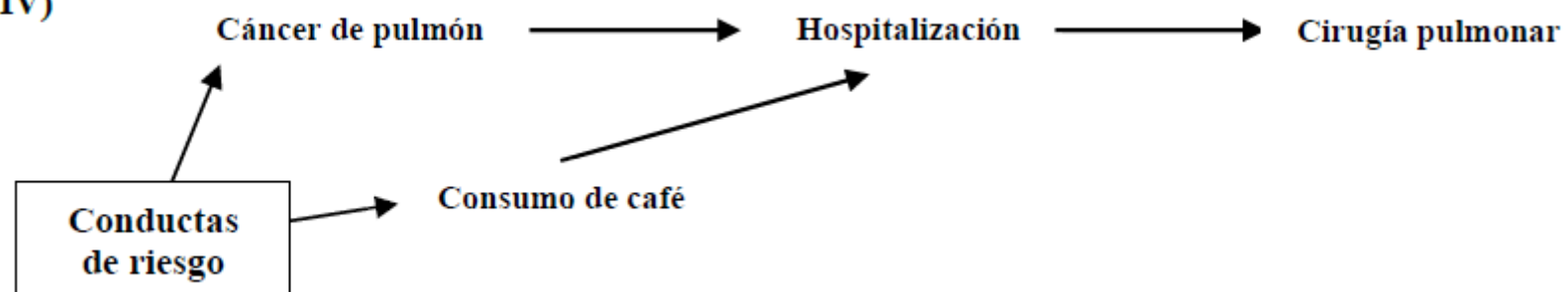


III)



* Estudio se realizó solamente en pacientes sometidos a cirugía pulmonar.

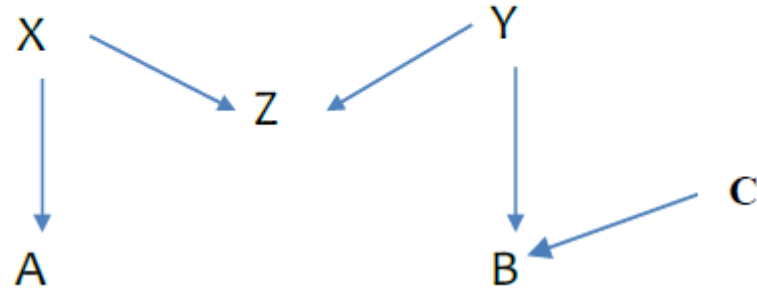
IV)



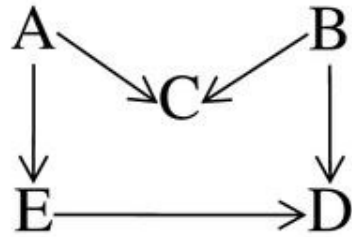
Responda verdadero (V) o falso (F) según corresponda.

- ☐ A.Existen solo dos criterios de Hill que no tienen excepciones al estimar causalidad: consistencia y temporalidad
- ☐ B. Una factor X puede ser denominado causa así no sea ni suficiente ni necesaria
- ☐ C.Usando métodos complejos de causalidad podemos estimar efectos individuales y poblacionales
- ☐ D.Los DAGs dan información sobre relaciones causales de una manera cualitativa y paramétrica.
- ☐ E.Los análisis de predicción son análisis de asociación sin implicar causalidad

Identifique en el siguiente DAG: i) las variables necesarias a ser ajustadas para obtener el efecto causal neto (sin confusión) de A en B; ii) ¿qué pasaría si se ajusta por la variable C?



Exercise: M-structure



1. Show the paths
2. Should we adjust for C?
3. If the design implies a selection on C, what would you call the resulting bias: selection bias or confounding?

1. Paths: (No adjusting)

$E \rightarrow D$

Causal

Open

$E \leftarrow A \rightarrow C \leftarrow B \rightarrow D$

Non-causal

Closed No bias

2. Paths: (Conditioning on C)

$E \rightarrow D$

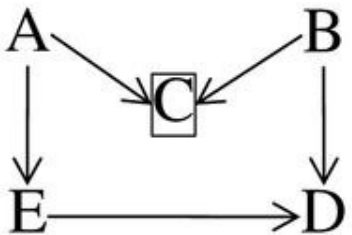
Causal

Open

$E \leftarrow A \rightarrow [C] \leftarrow B \rightarrow D$

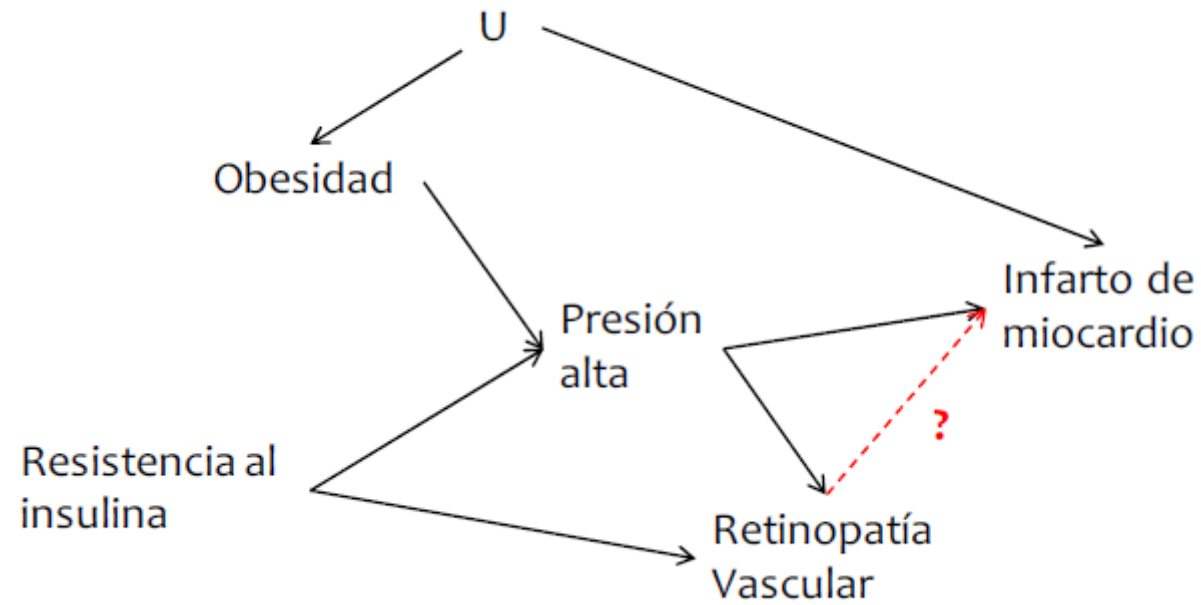
Non-causal

Open bias

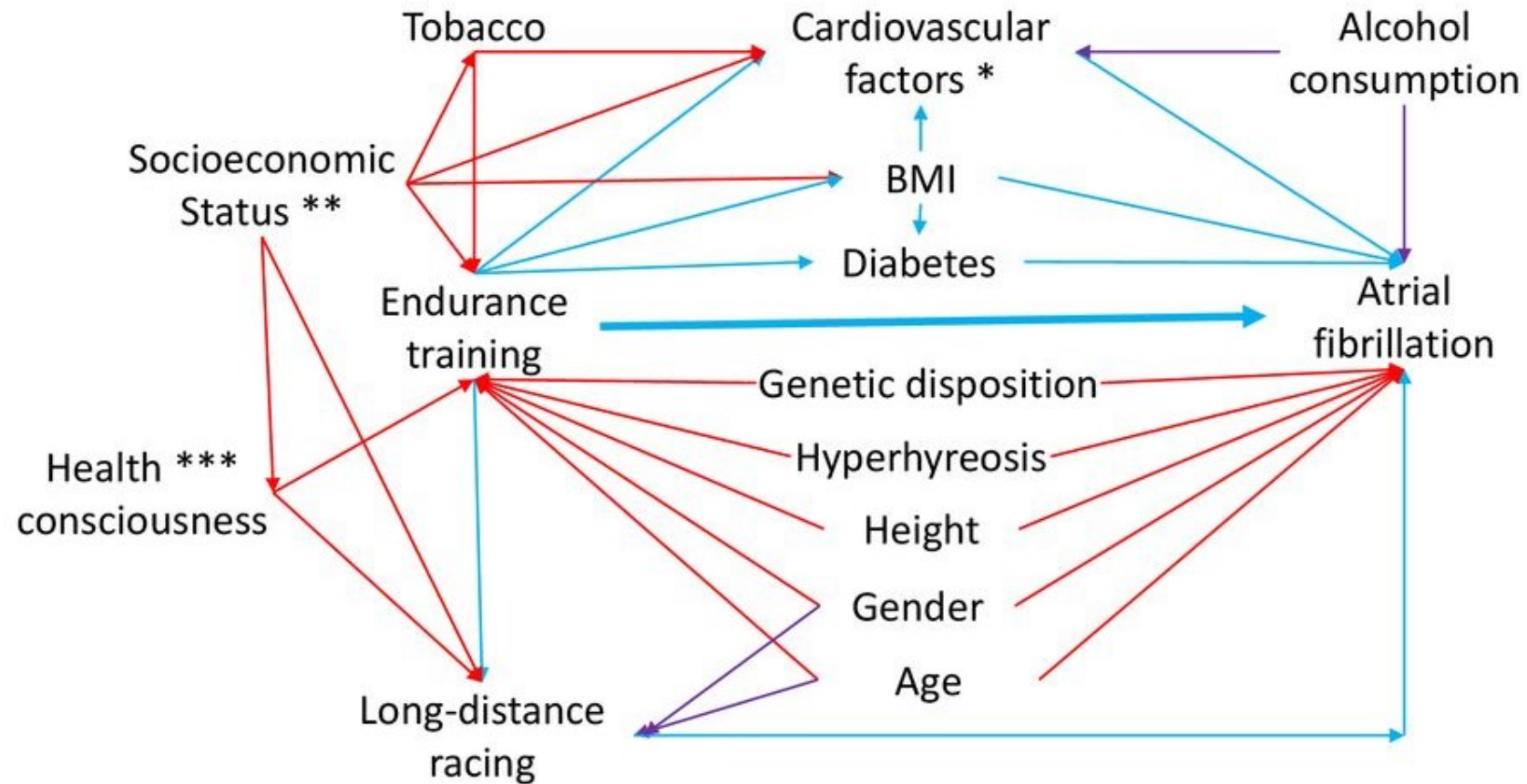


3. We are conditioning on a collider, hence: selection bias!

El siguiente DAG busca estimar el efecto **causal** de la retinopatía vascular en el infarto al miocardio:



Real world examples



Several arrows missing!

*Hypertension, heart disease, high cholesterol
** Socioeconomic status: Education, marital status
*** Unmeasured factors
(Blue: Mediators, red: confounders, violet: colliders)

Table 2 Fallacy

Not all coefficients are causal effects

Statistical model designed to identify $X \rightarrow Y$ will not also identify effects of control variables

Table 2 is dangerous

TABLE 2—ESTIMATED PROBIT MODELS
FOR THE USE OF A SCREEN

	Preliminaries blind		Finals blind
	(1)	(2)	(3)
(Proportion female) $_{t-1}$	2.744 (3.265) [0.006]	3.120 (3.271) [0.004]	0.490 (1.163) [0.011]
(Proportion of orchestra personnel with <6 years tenure) $_{t-1}$	−26.46 (7.314) [−0.058]	−28.13 (8.459) [−0.039]	−9.467 (2.787) [−0.207]
“Big Five” orchestra		0.367 (0.452) [0.001]	
pseudo R^2	0.178	0.193	0.050
Number of observations	294	294	434

Westreich & Greenland 2013 The Table 2 Fallacy

¡Gracias!

Causalidad: Parte IV

Resumen en 10 min



Daniel Fernández Guzmán, MD
danferguz@gmail.com

- › **Filiación:** UCSUR, EviSalud
- › **Cursó estudios en:** UNSAAC, UCC, UPCH, IECS, Univ Nebrija.
- › **Ha recibido pagos/financiamiento de:** UNSAAC, UCSUR, UPCH, MINSA (DIRESA Cusco), EviSalud, SOPECARD, IETSI
- › No tengo potenciales conflictos de interés con el tema de esta clase

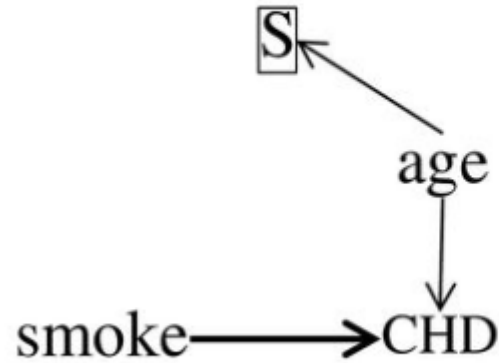
Selection bias

Selection bias: concept 1

Simple version

- “Selected different from unselected”
- Prevalence (D)
 - Old have lower prevalence than young
 - Old respond less to survey
 - ⇒ Selection bias: prevalence overestimated
- Effect ($E \rightarrow D$)
 - Old have lower effect of E than young
 - Old respond less to survey
 - ⇒ Selection bias: effect of E overestimated

“Selected different from unselected”



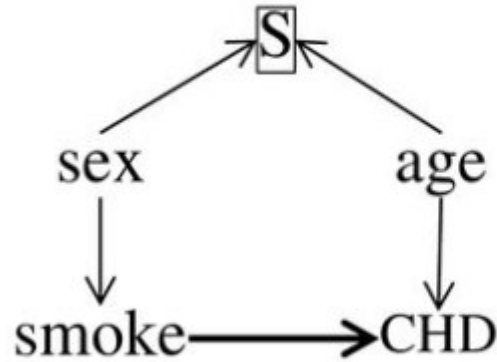
Paths (but see concept 3)			Type	Status
smoke→CHD			Causal	Open
Age	Population	RR _{smoke}	Selected	RR _{smoke}
Young	50 %	4.0	75 %	4.0
Old	50 %	2.0	25 %	2.0
All		3.0		3.5

Normally, selection variables unknown

Name:
interaction based?

- Properties:
 - Need smoke-age interaction
 - Cannot be adjusted for, but stratum effects OK
 - True RR=weighted average of stratum effects
 - RR in “natural” range (2.0-4.0)
 - Scale dependent

“Distorted E-D distributions”



Paths	Type	Status
smoke→CHD	Causal	Open
smoke←sex→[S]←age→CHD	Non-causal	Open

Properties:

Open non-causal path (collider)

Does not need interaction

Can be adjusted for (sex or age)

Not in “natural” range (“surprising bias”)

Name:

Collider stratification bias

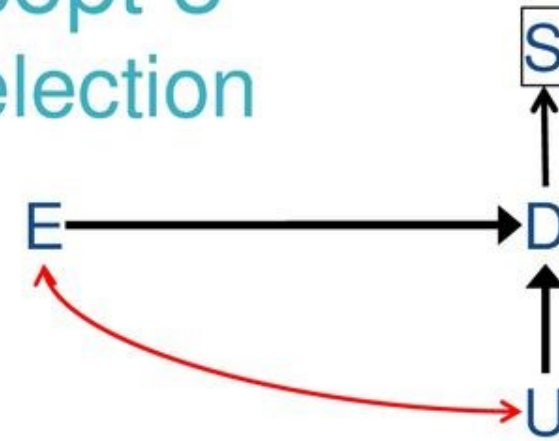
Selection bias types:

Berkson’s, loss to follow up, nonresponse, self-selection, healthy worker

Selection bias: concept 3

Outcome dependent selection

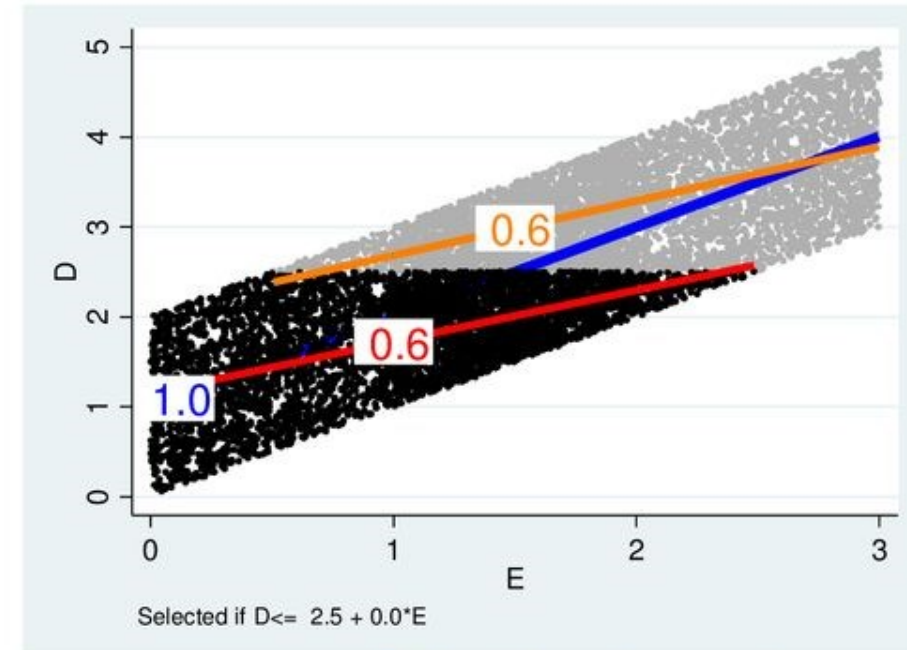
Selection into the study based on D.
Get bias among selected.



Explanation:

- Always have exogenous U.
- D is a collider on $E \rightarrow D \leftarrow U$, S is a descendant of collider D.
- Conditioning on (a descendant of) a collider opens the $E \rightarrow D \leftarrow U$ path, and U becomes associated with E.
- U now acts a confounder for $E \rightarrow D$.

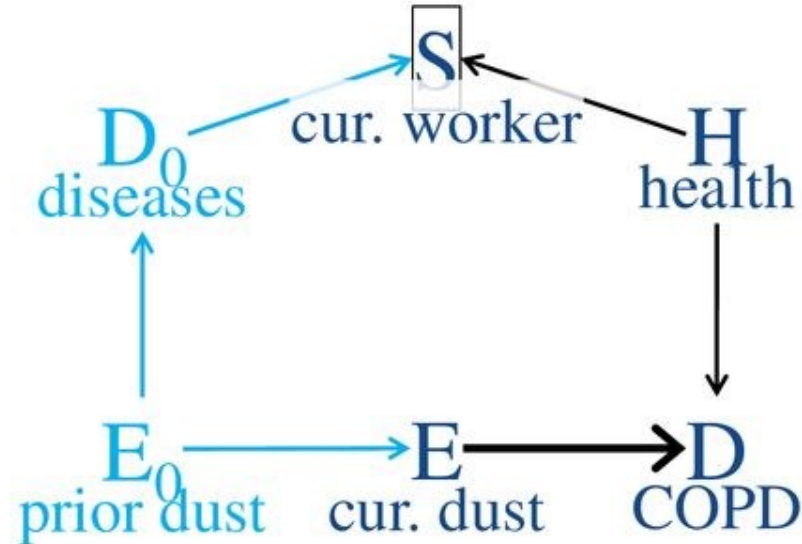
Example of **non-causal confounding**



Selection depends on:
Strength of $E \rightarrow D$. Strength of $U \rightarrow D$
Unmatched Case-Control

Exercise: Dust and COPD

Chronic Obstructive Pulmonary Disease

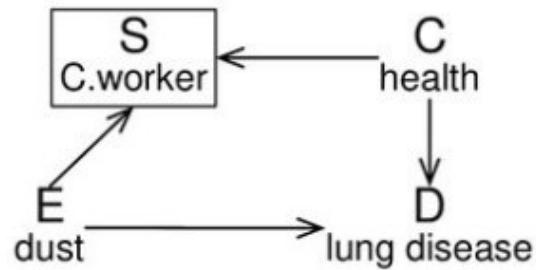


COPD risks:

		Dust	
		low	high
Health	good	5 %	10 %
	poor	10 %	20 %

1. Calculate the RR of dust on COPD in good and poor health groups.
2. Write down the paths for the effect of E on D. E_0 and D_0 are unknown (past) measures.
3. What would you adjust for?
4. Suppose the crude effect of dust on COPD is $RR=0.7$ and the true $RR=2$. What do you call this bias?
5. Could the concept 1 (interaction based) selection bias work here?

Dust and COPD



Paths:

$E \rightarrow D$

Causal

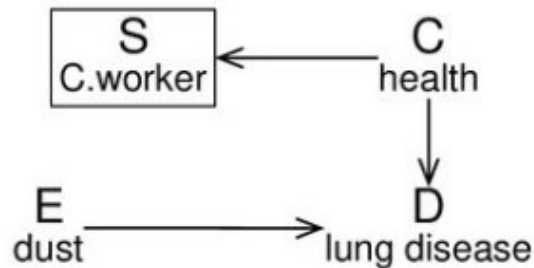
Open

$E \rightarrow [S] \leftarrow C \rightarrow D$

Non-causal

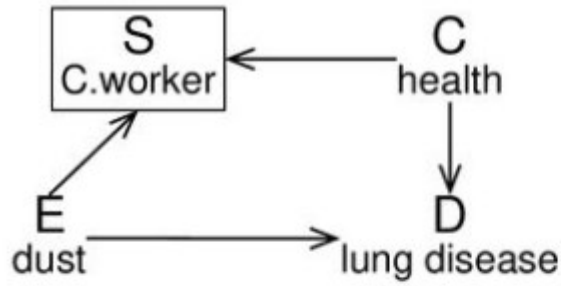
Open

- Selection bias (**collider stratification**)
- Healthy worker effect, can show protective effects

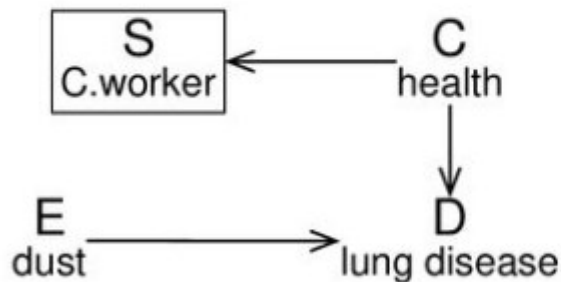


- Selection bias (**interaction based**)
 - Good health: $RR_{ED} = 2.0$
 - Bad health: $RR_{ED} = 3.0$
 Interaction
- Can not show protective effect

Dust and COPD

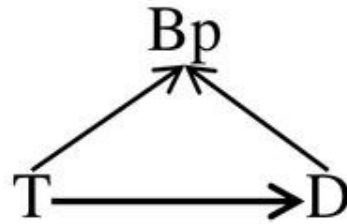


- Selection bias (**collider stratification**)
- Healthy worker effect, can show protective effect
- Can show untrue effects
- Adjust for health to remove bias



- Selection bias (**interaction based**)
Good health: $RR_{ED} = 1.0$
Bad health: $RR_{ED} = 2.0$ } Interaction
- Can not show protective effect
- The E-D arrow is not required
- Can only show “true effects”

Exercise: Simpson's paradox



Want the effect of treatment (T) on disease (D)
Both T and D affect blood pressure (Bp)

1. Draw the DAG
2. Calculate the population effect of T on D
3. Conclusions?

Population

		D		sum	risk
		1	0		
T	1	70	130	200	0.4
	0	230	570	800	0.3
				1000	

$$RR = 1.2$$

20% more disease if treated

Unbiased

Low blood pressure

		D		sum	risk
		1	0		
T	1	65	45	110	0.6
	0	180	90	270	0.7
				380	

$$RR = 0.9$$

10% less disease if treated

Collider bias

High blood pressure

		D		sum	risk
		1	0		
T	1	5	85	90	0.06
	0	50	480	530	0.09
				620	

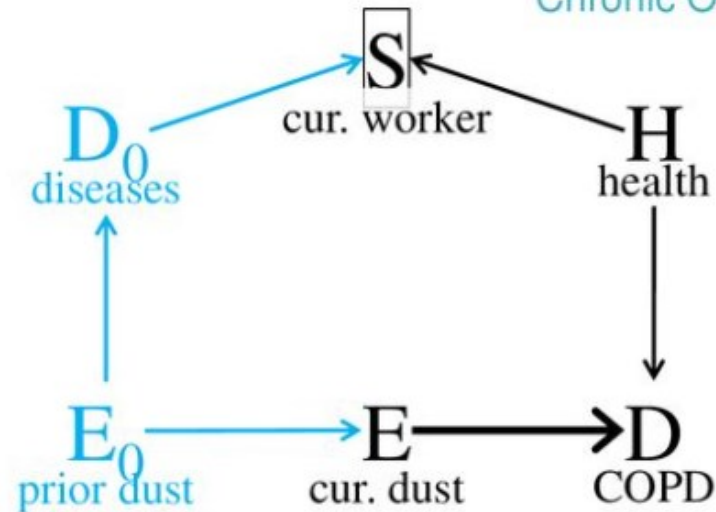
$$RR = 0.6$$

40% less disease if treated

Collider bias

Dust and COPD

Chronic Obstructive Pulmonary Disease



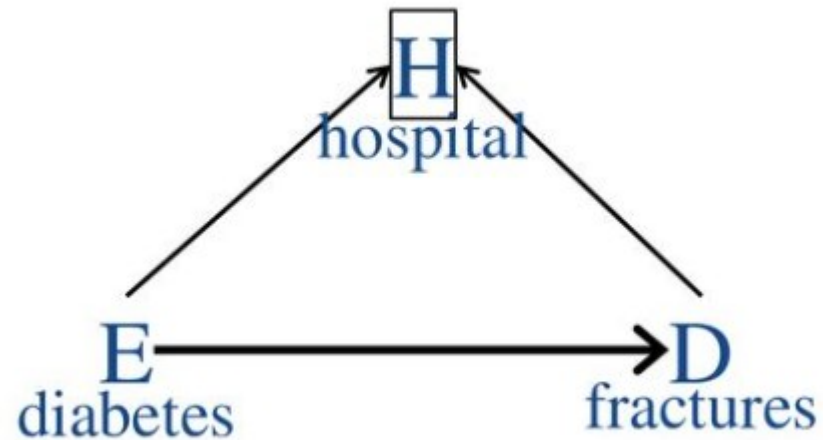
COPD risks:

		Dust			
		low	high	RR_{dust}	RD_{dust}
Health	good	5 %	10 %	2.00	5 %
	poor	10 %	20 %	2.00	10 %

1. No interaction, can we still have selection bias?
2. Paths:

$E \rightarrow D$	Causal	Open
$E \leftarrow E_0 \rightarrow D_0 \rightarrow [S] \leftarrow H \rightarrow D$	Non-causal	Open
3. Crude $RR=0.7$. Adjust for H , Health: True $RR=2$.
4. Healthy worker bias (selection bias concept 2)
5. No interaction based selection bias, no interaction (on RR scale, but on the RD).

Convenience sample, homogenous sample



1. **Convenience:**
Conduct the study among
hospital patients?

2. **Homogeneous sample:**
Population data,
exclude hospital patients?

Conditional

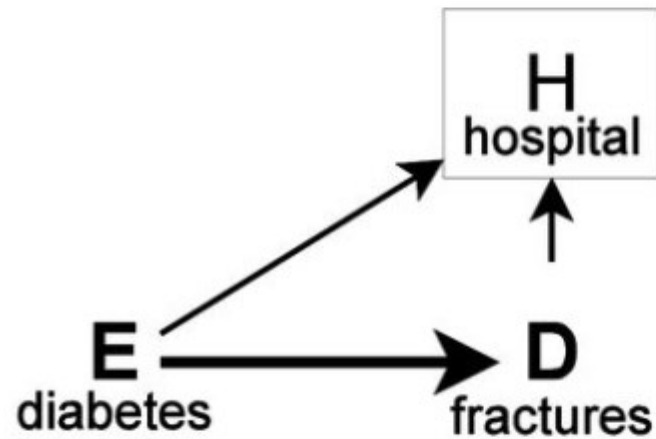
	Path	Type	Status
1	$E \rightarrow D$	Causal	Open
2	$E \rightarrow [H] \leftarrow D$	Non-Causal	Open

} Collider, selection bias

Collider stratification bias: at least on stratum is biased

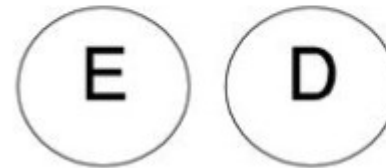
Exercise: causal pies

1. Causal pies for hospital
2. Selection bias:



Question
Variable with only 1 value?

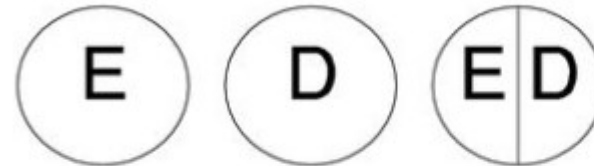
Sufficient causes for Hospital:



1) or



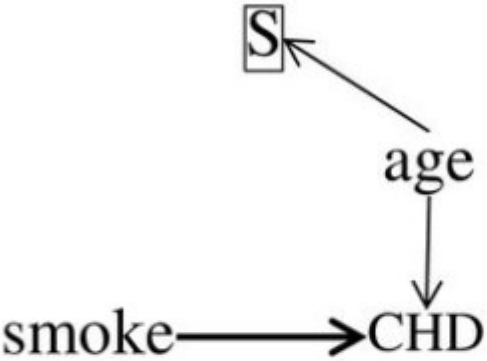
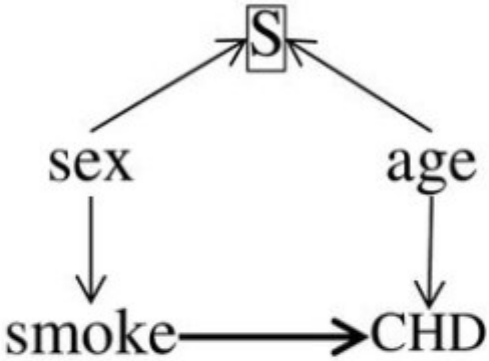
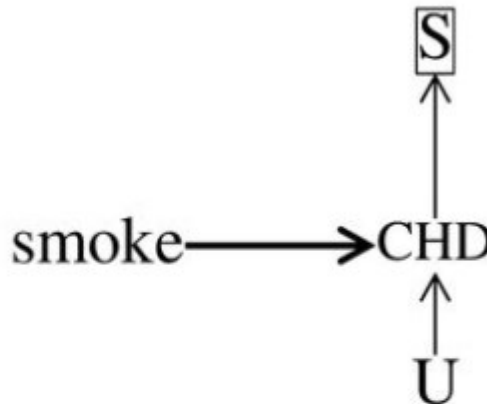
2) and



3) both

Selection bias:

- 1) Negative bias
- 2) Positive bias
- 3) ?

Concept 1	Concept 2	Concept 3
 <pre> graph TD S[S] --> age age --> CHD smoke --> CHD </pre>	 <pre> graph TD S[S] --> sex S[S] --> age sex --> smoke age --> CHD smoke --> CHD </pre>	 <pre> graph TD smoke --> CHD U --> CHD CHD --> S[S] </pre>
Selected differ from unselected in E-D effects	Selected differ from unselected in E-D distributions	Selected differ from unselected in E-D distributions
Interaction based selection	Collider stratification bias	Outcome dependent selection
"natural" effects	"surprising" effects	variance dependent
Report stratum effects	Adjust	IPW

Time Dependent Confounding

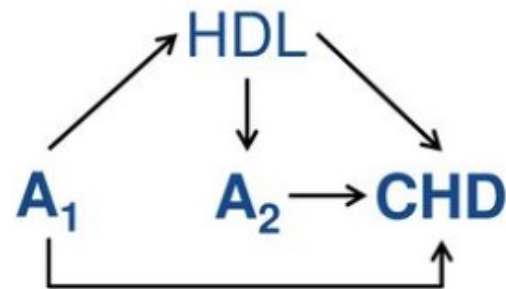
Motivating example

Population:

Exposure: Alcohol use at two time points

Mediator/confounder: HDL cholesterol

Outcome: Coronary Heart Disease



Time Dependent Confounder:
a **confounder (HDL)**
that depends on
earlier exposure (A_1)

Estimate: Joint effect of alcohol use (or effect of A_1 and A_2)

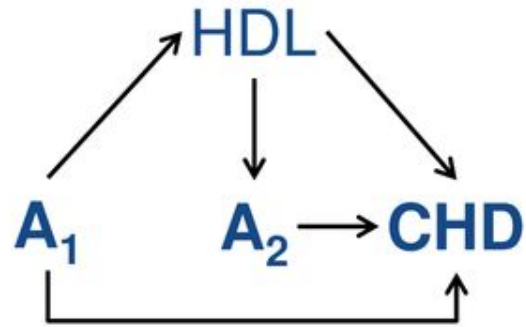
Simplified DAG, several variables and arrows missing

Could have HDL_1 and HDL_2

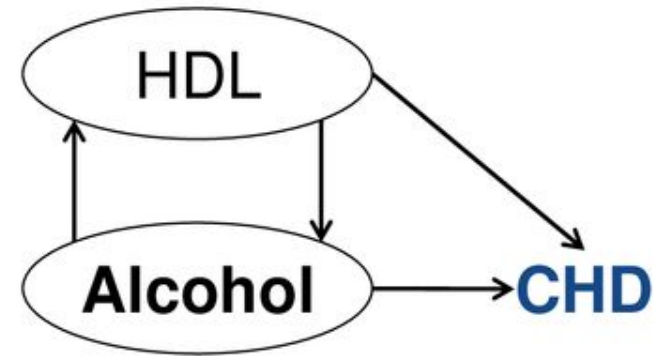
Common situation in patient-doctor follow up!

Alcohol and CHD

DAG



Process graph



HDL is a **confounder**

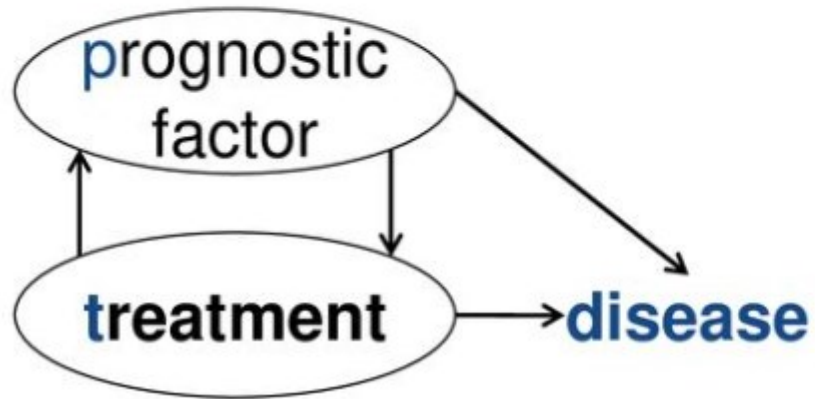
HDL is a **mediator**

The process graph is **simpler** but less **specific**

Ordinary adjustment does not work

Must use Inverse Probability of Treatment Weighting, IPTW

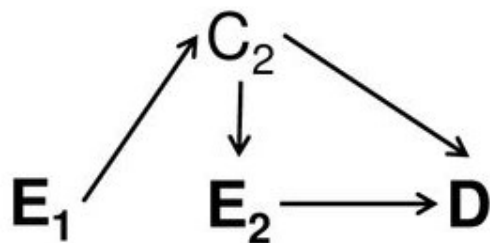
Patient-Doctor



Treatment regulated by level of prognostic factor. Both affect later disease.

Exercise: Time Dependent Confounding, Variants

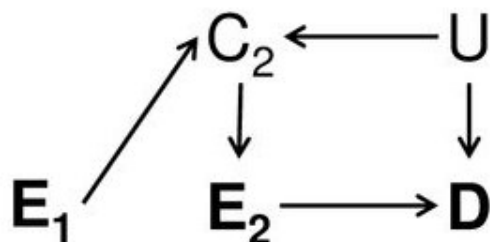
Variant 1:



Variant 1:

- a) Show the paths from E_1 to D
- b) Show the paths from E_2 to D
- c) Can you estimate the joint effect ($E_1 + E_2$) in one ordinary regression model?

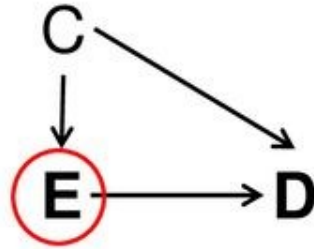
Variant 2:



Variant 2:

If time, do the same for variant 2

Inverse Probability of Treatment Weighting



Simple point treatment (exposure)

Focus on probability of being exposed (binary)

Subjects

		E		
		1	0	sum
C	1	300	100	400
	0	200	400	600
				1000

Probabilities

		E		
		$P(E)$	$P(\bar{E})$	sum
C	1	0.75	0.25	1
	0	0.33	0.67	1

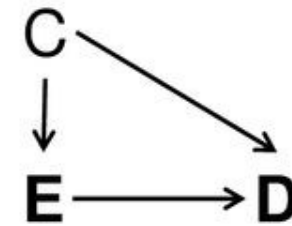
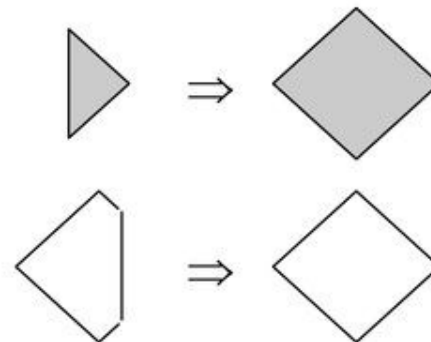
Propensity scores

Weights

		E		
		$1/P(E)$	$1/P(\bar{E})$	
C	1	1.3	4.0	
	0	3.0	1.5	

"Subjects"

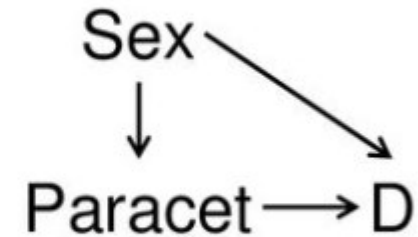
		E		
		N^*w	N^*w	sum
C	1	400	400	800
	0	600	600	1200
		1001	1000	2000



Weighted analysis!

Inverse Probability of Treatment Weighting, Exercise

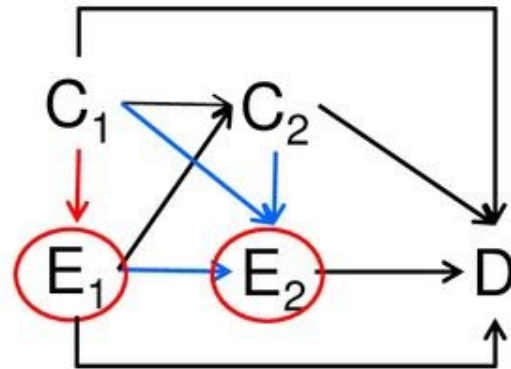
Sample: 200 females, 100 use Paracet
 800 males, 200 use Paracet



1. Calculate the risk of Paracetet use for each sex.
2. Calculate the RR of Paracetet use for females versus males
3. Do an inverse probability of treatment weighting for Paracetet.
4. Calculate the RR of Paracetet use for females versus males in the reweighted pseudo data
5. Explain the results in the DAG

Inverse Probability of Treatment Weighting

Time varying treatment (exposure)



E is treatment, D is disease
C is a prognostic factor

Time points t_1 and t_2

Weights:

w_1

w_2

Weight at E_2 :

Weight for the entire exposure and covariate history up to time 2

Weight by $w_1 * w_2$

Ordinary weights:

$$w(t) = \prod_{k=1}^t \frac{1}{P(\text{exposure at } k | \text{cov. and exp. up to } k)}$$

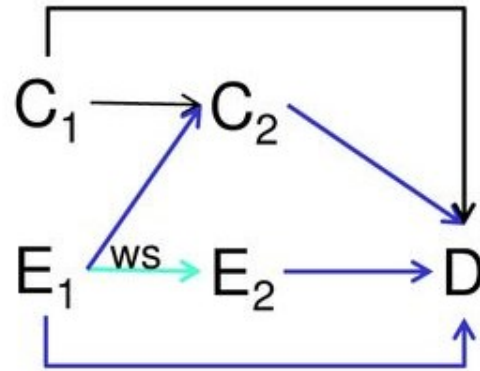
Vary a lot

Stabilized weights:

$$ws(t) = \prod_{k=1}^t \frac{P(\text{exposure at } k | \text{exposure up to } k)}{P(\text{exposure at } k | \text{cov. and exp. up to } k)}$$

Vary less

M_{arginal} S_{tructural} M_{odel}



DAG for the reweighted pseudo data

MSM:

The expected value of a counterfactual outcome D
under a hypothetical exposure $\bar{e}=(e_1,e_2)$:

$$E(D^{\bar{e}}) = \alpha_0 + \alpha_1 e_1 + \alpha_2 e_2$$

$$\text{Joint effect} = E(D^{\bar{1}}) - E(D^{\bar{0}})$$

MSM in Stata

* Probability of E1

```
regress E1 C1
```

```
predict p1
```

```
replace p1=1-p1 if E1==0
```

* Probability of E2

```
regress E2 E1 C1 C2
```

```
predict p2
```

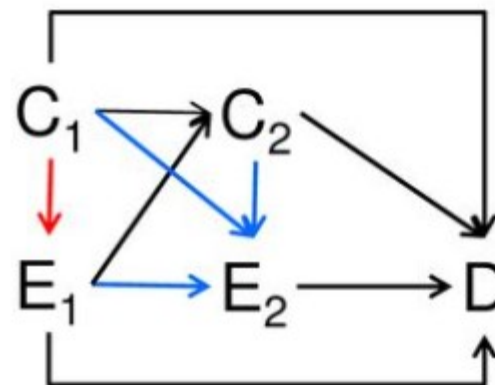
```
replace p2=1-p2 if E2==0
```

* Weights

```
generate w=1/(p1*p2)
```

* MSM

```
regress D E1 E2 [pw=w]
```

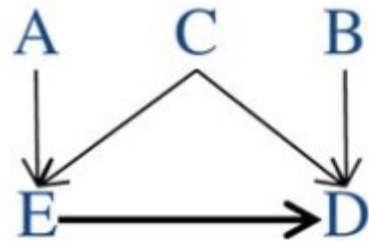


$$w(t) = \prod_{k=1}^t \frac{1}{P(\text{exposure at } k | \text{cov. and exp. up to } k)}$$

$$E(D^{\bar{e}}) = \alpha_0 + \alpha_1 e_1 + \alpha_2 e_2$$

Easy-peasy!

Effects of adjustment

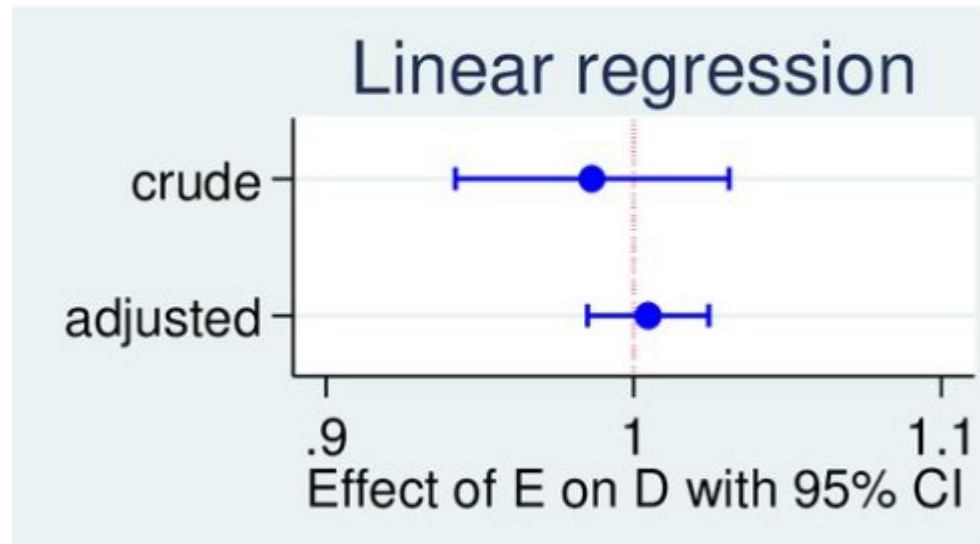
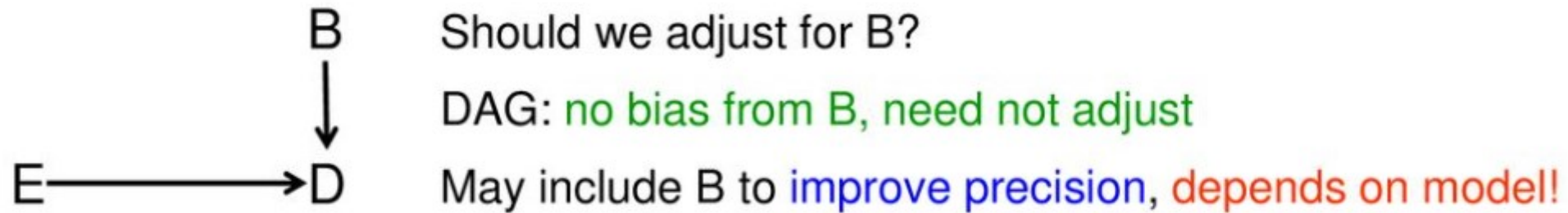


What variables should we adjust for?

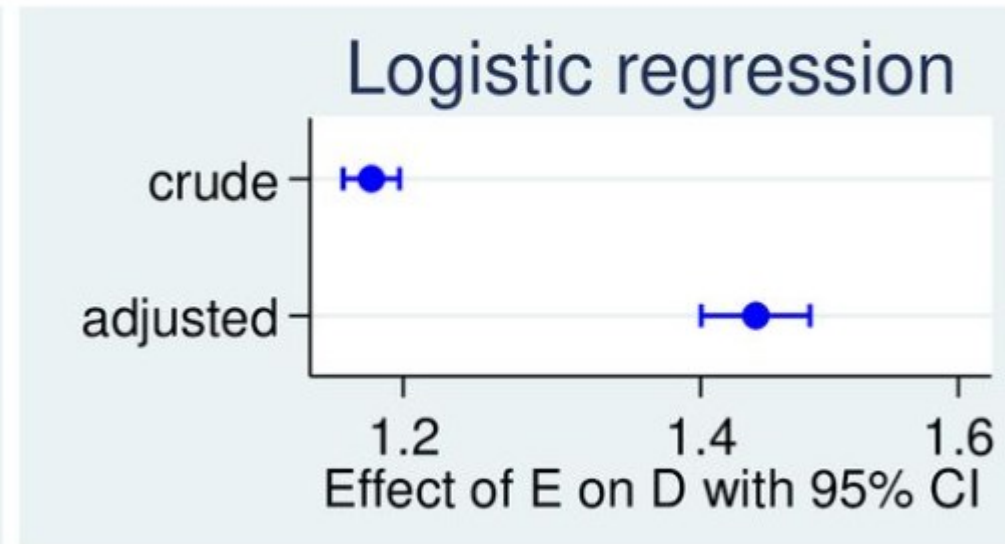
What are the effects of adjustment?

Variable	Adjust	Bias	Precision
A	no	bias amplification	reduce precision (collinearity)
B	maybe	no	improve precision (model dependent)
C	yes	remove confounding	?

Effects of adjustment: Precision

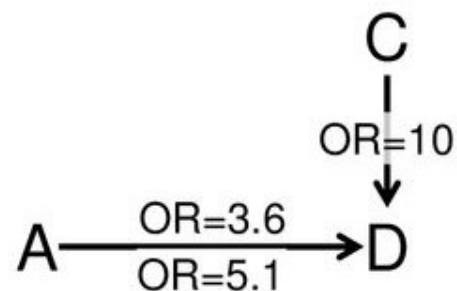


Including B: better precision



Including B: worse precision
OR not collapsible

Non-collapsibility of the OR



Population

		D		sum	odds
		1	0		
A	1	470	530	1 000	0.89
	0	1 775	7 225	9 000	0.25
				10 000	
		OR=		3.6	

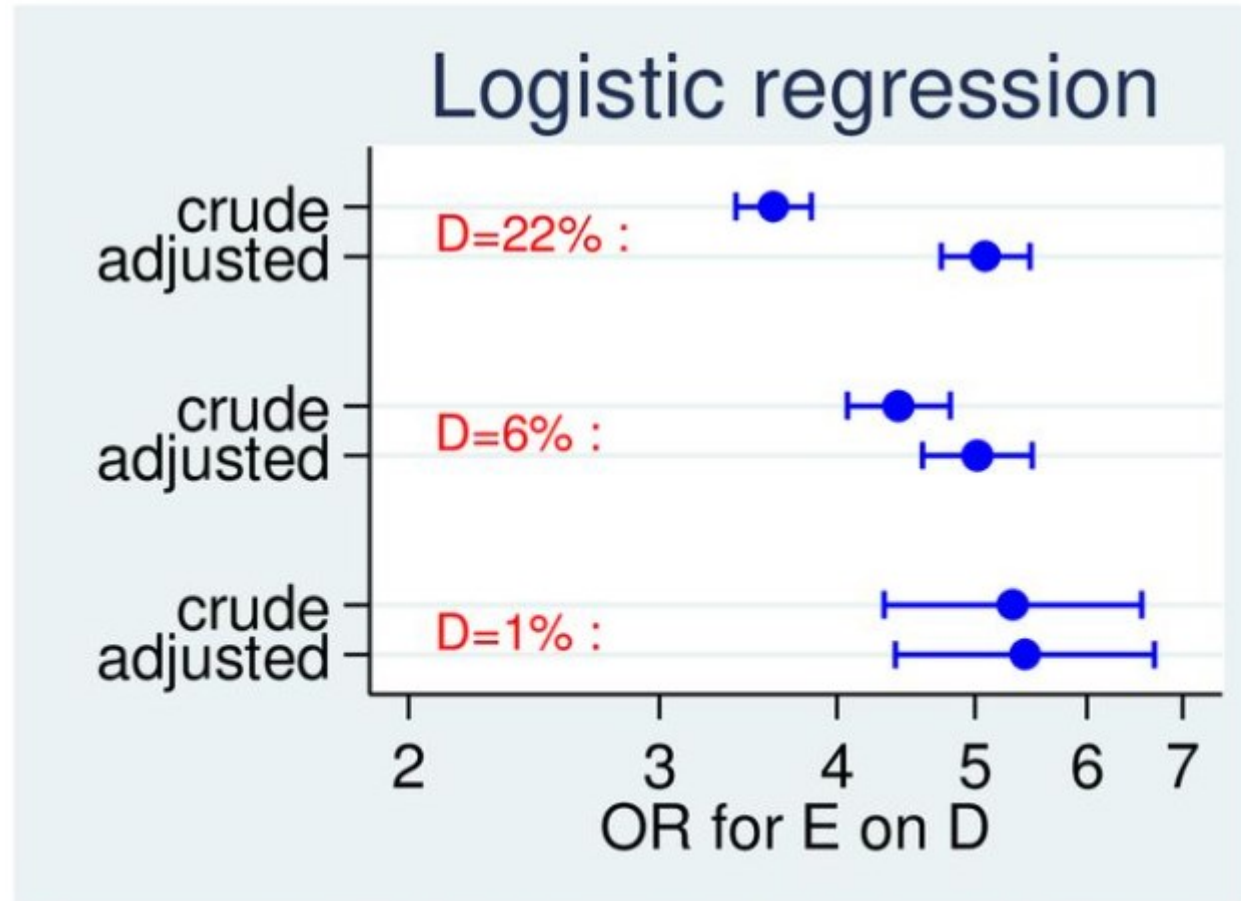
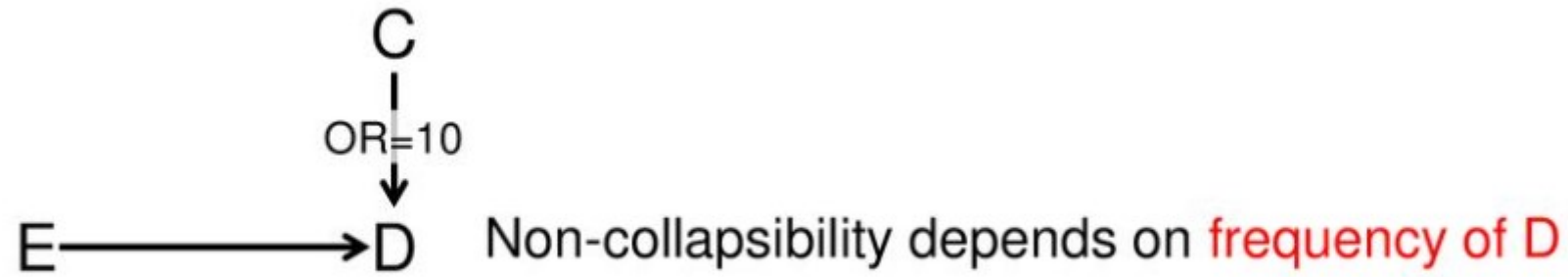
C=0

		D		sum	odds
		1	0		
A	1	105	395	500	0.27
	0	225	4 275	4 500	0.05
				5 000	
		OR=		5.1	

C=1

		D		sum	odds
		1	0		
A	1	365	135	500	2.70
	0	1 550	2 950	4 500	0.53
				5 000	
		OR=		5.1	

Non-collapsibility of the OR

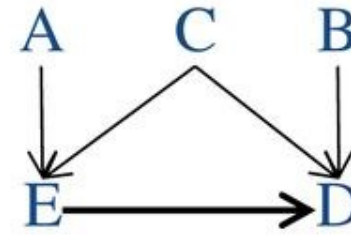


Not collapsible

Appr. collapsible

Collapsible

Summing up so far



- Adjustment
 - Should not adjust for A (reduce precision)
 - May adjust for B (improve precision)
 - Should adjust for C (remove confounding)
- Collapsibility
 - Collapsible measures:
 - Risk Difference (RD), Rate Difference, Risk Ratio (RR)
 - Non-collapsible measures:
 - Odds Ratio (OR), Rate Ratio (HRR)
 - Adjusting for B changes OR (and the HRR) but not due to confounding

¡Gracias!

Causalidad: Parte V

Resumen en 10 min



Daniel Fernández Guzmán, MD
danferguz@gmail.com

- › **Filiación:** UCSUR, EviSalud
- › **Cursó estudios en:** UNSAAC, UCC, UPCH, IECS, Univ Nebrija.
- › **Ha recibido pagos/financiamiento de:** UNSAAC, UCSUR, UPCH, MINSA (DIRESA Cusco), EviSalud, SOPECARD, IETSI
- › No tengo potenciales conflictos de interés con el tema de esta clase

METHODS TO REMOVE CONFOUNDING

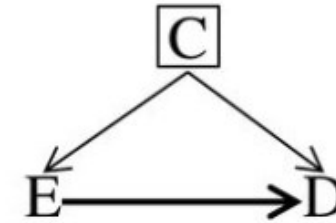
Method

Action

DAG effect

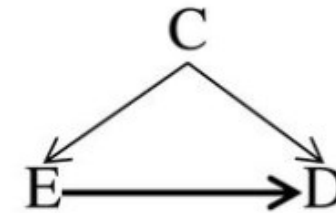
Condition: Restrict, Stratify, Adjust

Close path



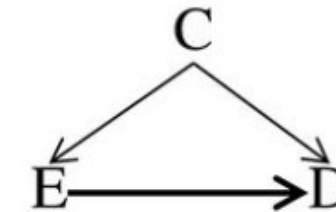
Cohort matching, Propensity Score
Inverse Probability Treatment Weighting

Remove CE arrow



~~Case Control matching?~~
Other methods?

Remove CD arrow

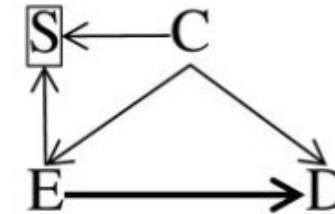


Matching: Cohort vs Case-Control

Matching in cohort, binary E

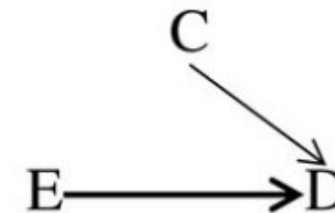
Matching:

For every **exposed** person with a **value of C**,
find an **unexposed** person with the **same value of C**



- ⇒ selected based on E and C
- ⇒ E independent of C after matching
- ⇒ All open paths between C and E must

$$\left. \begin{array}{l} C \rightarrow E \\ C \rightarrow [S] \leftarrow E \end{array} \right\} \text{sum to "null"}$$



Unfaithful DAG

Cohort matching removes confounding

Cohort matching is not common,
except in propensity score matching

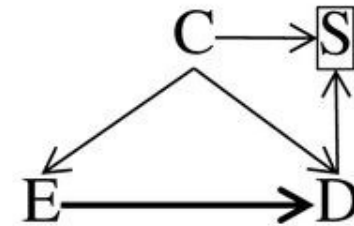
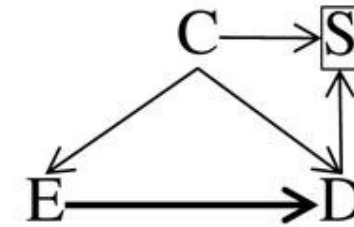
Matching in Case Control, binary D

Matching:

For every **case** with a **value of C**,
find a **control** with the **same value of C**

- ⇒ selected based on D and C
- ⇒ D independent of C after matching
- ⇒ All open paths between C and D must

$$\left. \begin{array}{l} C \rightarrow D \\ C \rightarrow [S] \leftarrow D \\ C \rightarrow E \rightarrow D \end{array} \right\} \text{sum to "null"}$$



Case-Control matching does not remove confounding,
unless $E \rightarrow D = 0$ (or $C \rightarrow E = 0$)

⇒ must adjust for C in all analyses

Case-Control matching common, may improve precision

Drawing a causal DAG

Start: E and D

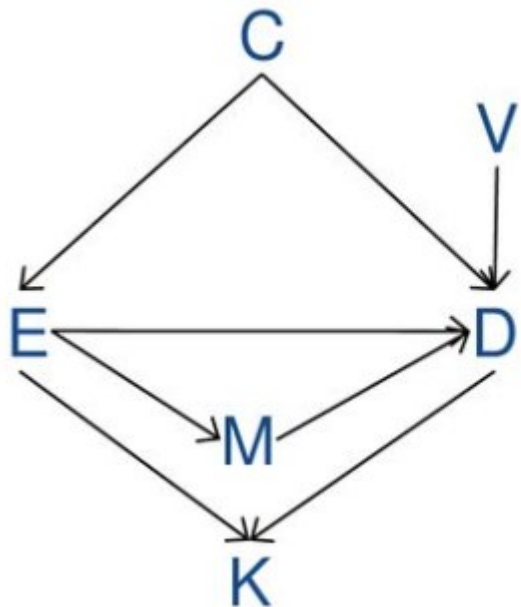
add: [S]

add: C-s

1 exposure, 1 disease

variables conditioned by design

all common causes of 2 or more
variables in the DAG



C **must** be included
V may be excluded
M may be excluded
K may be excluded

common cause
exogenous
mediator
unless we condition

Exercise: Drawing survivor bias

1. We want to study the effect of exposure early in life (E) on disease (D) later in life.
2. Exposure (E) decreases survival (S) in the period before D (deaths from other causes than D).
3. A risk factor (R) reduces survival (S) in the period before D.
4. The risk factor (R) increases disease (D).
5. *Only survivors are available for analysis* (look at Collider idea).

Draw and analyze the DAG

DAGitty background

- DAGitty
 - Draw DAGs
 - Analyze DAGs
 - Test DAGs
- Web page
 - <http://www.dagitty.net/>
 - Run or download

Interface

Diagram style

classic

SEM-like

View mode

normal

moral graph

correlation graph

Coloring

☒ causal paths

☒ biasing paths

☒ ancestral structure

Effect analysis

☐ atomic direct effects

Legend

exposure

outcome

ancestor of exposure

Model

Examples

How to ...

Layout

Help

```
graph TD; Age((Age)) --> Obesity((Obesity)); Age((Age)) --> Vitamin((Vitamin)); Vitamin((Vitamin)) --> BirthDefects((Birth defects)); Obesity((Obesity)) --> BirthDefects((Birth defects)); Vitamin((Vitamin)) --> BirthDefects((Birth defects));
```

Causal effect identification

Adjustment (total effect)

Minimal sufficient adjustment sets for estimating the total effect of Vitamin on Birth defects:

- Age
- Obesity

Testable implications

The model implies the following conditional independences:

- Age $\perp$ Birth defects | Obesity, Vitamin
- Obesity $\perp$ Vitamin | Age

[Export R code](#)

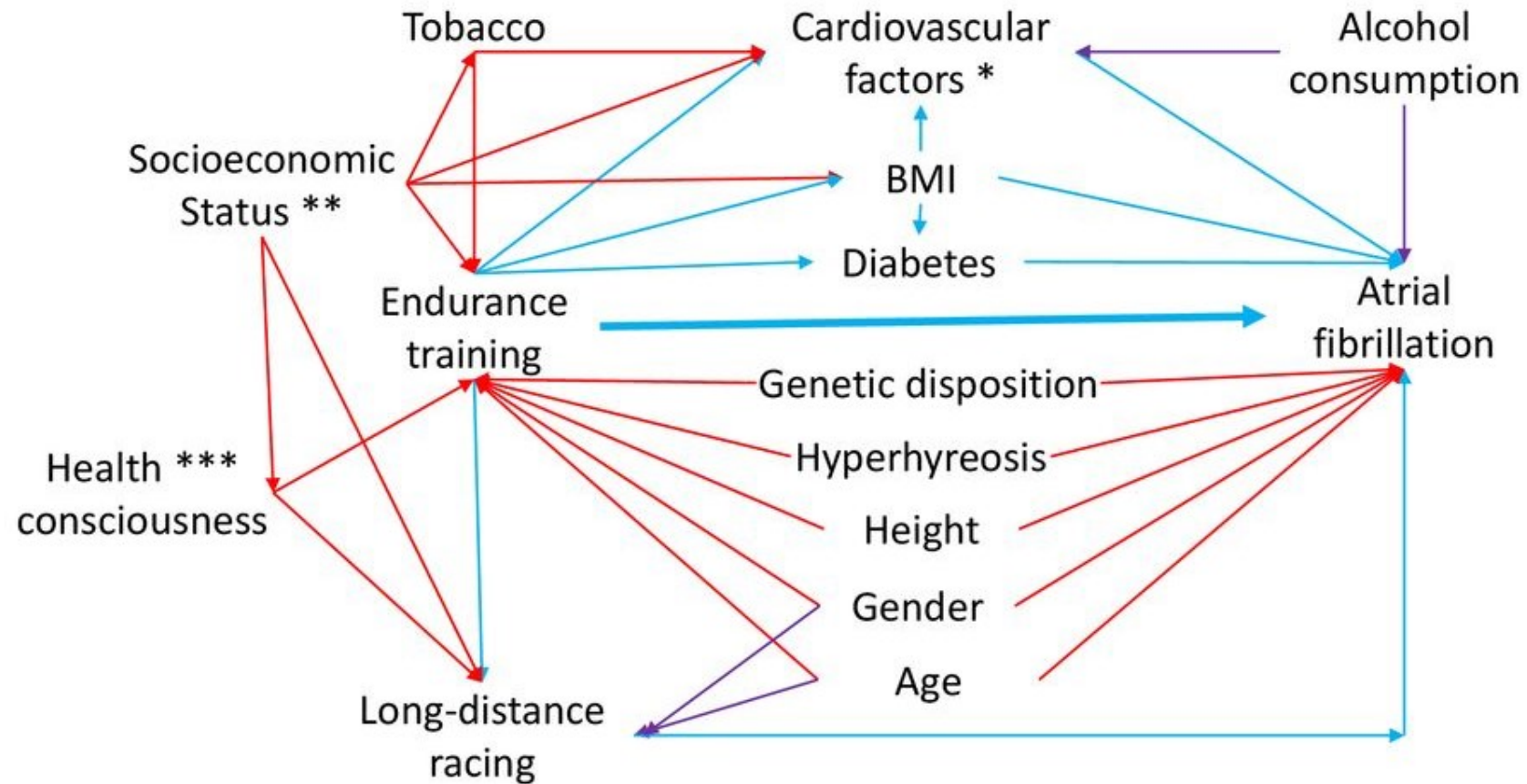
Model code

Age 1 @0.100,0.800
Birth%20defects 0 @0.500,1.000
Obesity 1 @0.500,0.800
Vitamin E @0.100,1.000

Age Obesity Vitamin

- Draw new model
 - Model>New model, Exposure, Outcome
- New variables, connect, rename
 - n new variable (or double click)
 - c connect (hit c over V1 and over V2 to connect)
 - r rename
 - d delete
- Status (toggle on/off)
 - e exposure
 - o outcome
 - u unobserved
 - a adjusted

Real world examples



Several arrows missing!

*Hypertension, heart disease, high cholesterol
** Socioeconomic status: Education, marital status
*** Unmeasured factors
(Blue: Mediators, red: confounders, violet: colliders)

Randomized experiments Mendelian randomization

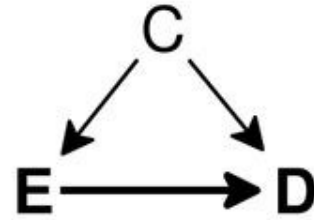
Strength of arrow, randomization



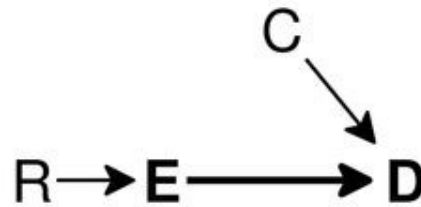
Sub analysis conditioning on E
may lead to bias

Path	Type	Status
1 $R \rightarrow E \rightarrow D$	Causal	open
2 $R \rightarrow E \leftarrow U \rightarrow D$	non-causal	closed

Randomized experiments



Observational study

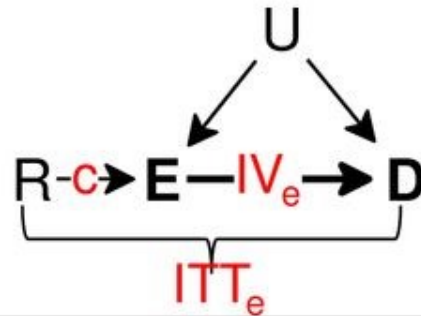


Randomized experiment with full compliance

R = randomized treatment

E = actual treatment.

$R=E$



Randomized experiment with less than full compliance (c)

If linear model: $ITT_e = c * IV_e$, $c < 1$

Intention To Treat effect:

Per Protocol:

Instrumental Variable effect:

crude

adjusted

effect of R on D (unconfounded)

effect of E on D (confounded by U)

effect of E on D (if c is known, 2SLS)

population

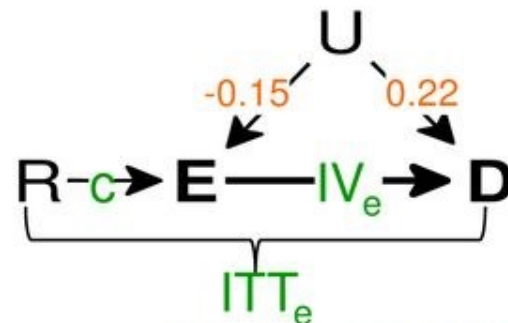
individual

RCT exercise

		E		N	Risk
		+	-		
R	+	85	15	100	0.85
	-	0	100	100	0.00
		85	115	200	

		D		N	Risk
		+	-		
R	+	43	57	100	0.43
	-	63	37	100	0.63
		106	94		

		D		N	Risk
		+	-		
E	+	32	53	85	0.38
	-	74	41	115	0.64
		106	94		



R+ means randomized to treatment,
 E+ means treated and D+ means getting disease.
 0.85 is the risk of treatment for R+ subjects,
 0.00 is the risk for R- subjects,
 the risk difference is the difference between these.

1. Calculate the compliance (c) as a risk difference from the table.
2. Calculate the intention to treat effect (ITT_e) as a risk difference.
3. Calculate the per-protocol effect (PP) as a risk difference.
4. Calculate the instrumental variable effect (IV_e).
5. Explain the results in words.

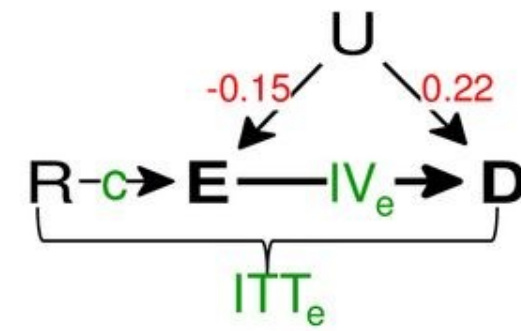
10 minutes

RCT

		E		N	Risk	RD	
		+	-				
R	+	85	15	100	0.85	0.85	c compliance
	-	0	100	100	0.00	0	
		85	115	200			

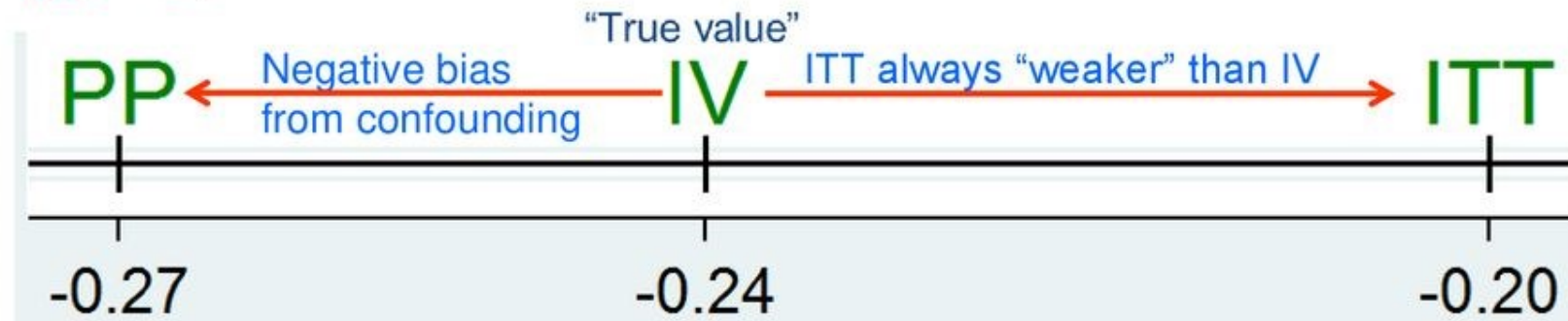
		D		N	Risk	RD	
		+	-				
R	+	43	57	100	0.43	-0.20	ITT ITT effect
	-	63	37	100	0.63	0	
		106	94				

		D		N	Risk	RD	
		+	-				
E	+	32	53	85	0.38	-0.27	PP PP effect
	-	74	41	115	0.64	0	
		106	94				



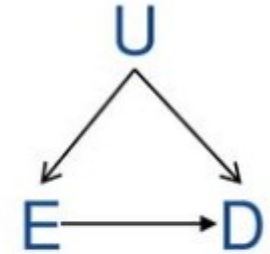
$$IV = \frac{ITT}{c} = \frac{-0.2}{0.85} = -0.24$$

Stata command for IV:
ivregress (2SLS)



Mendelian randomization

- Observational study
 - Suffers from unmeasured confounding



- Randomized trial: 3 conditions

1. R affects E:

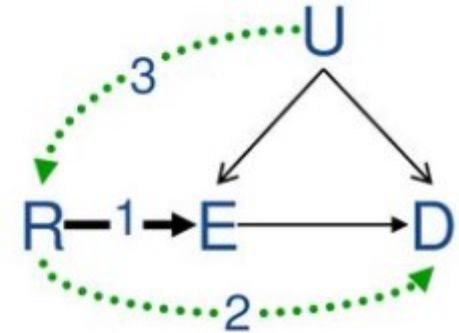
balanced, strong effect

2. No direct R-D effect:

R independent of D | E

3. R and D no common causes:

R independent of U



- Mendelian randomization: 3 conditions

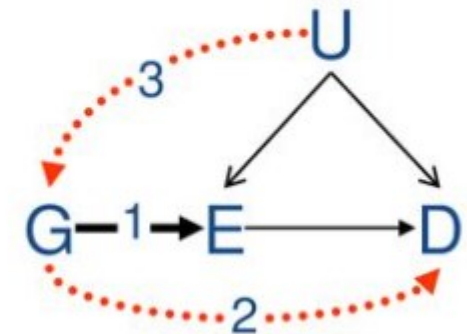
1. G must affect E:

unbalanced, weak $\Rightarrow$ large N

2. No direct G-D effect:

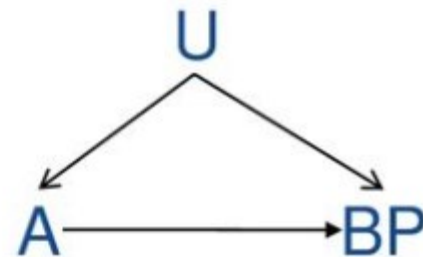
depends on gene function

3. G and D no common causes: Mendel's 2. law

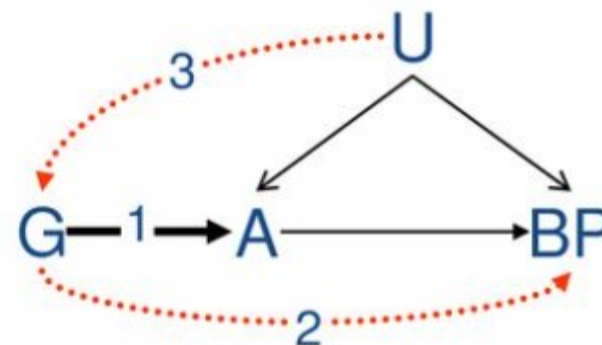
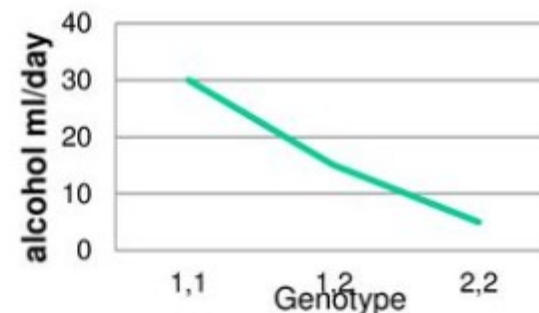


Ex: Alcohol and blood pressure

- Observational study
 - Alcohol use increases blood pressure
 - Many "lifestyle" confounders
- Gene: ALDH2, 2 alleles
 - 2,2 type suffer nausea, headache after alcohol
 - $\Rightarrow$ low alcohol regardless of lifestyle (U)
- Medelian randomization
 1. Gene ALDH2 is **highly** associated with alcohol
 2. **OK**, gene function is known
 3. **Mendel's 2. law**, no ass. to obs. confounders
- Result:
 - 1,1 type BP +7.4 mmHg
 - Alcohol increases blood pressure



Alcohol use



Summing up

- Data driven analyses **do not work**. Need **causal** information from outside the data.
- DAGs are **intuitive** and **accurate** tools to display that information.
- Paths show the flow of **causality** and of **bias** and guide the analysis.
- DAGs **clarify** concepts like confounding and selection bias, and show that we can adjust for both.

Better discussion based on DAGs

**Draw your assumptions
before your conclusions**

- Books

- Hernan, M. A. and J. Robins. *Causal Inference*. Web:
- Rothman, K. J., S. Greenland, and T. L. Lash. *Modern Epidemiology*, 2008.
- Morgan and Winship, *Counterfactuals and Causal Inference*, 2009
- Pearl J, *Causality – Models, Reasoning and Inference*, 2009
- Veierød, M.B., Lydersen, S. Laake, P. Medical Statistics. 2012

- Papers

- Greenland, S., J. Pearl, and J. M. Robins. *Causal diagrams for epidemiologic research*, Epidemiology 1999
- Hernandez-Diaz, S., E. F. Schisterman, and M. A. Hernan. *The birth weight "paradox" uncovered?* Am J Epidemiol 2006
- Hernan, M. A., S. Hernandez-Diaz, and J. M. Robins. *A structural approach to selection bias*, Epidemiology 2004
- Berk, R.A. *An introduction to selection bias in sociological data*, Am Soc R 1983
- Greenland, S. and B. Brumback. *An overview of relations among causal modeling methods*, Int J Epidemiol 2002
- Weinberg, C. R. *Can DAGs clarify effect modification?* Epidemiology 2007

- Aalen OO, Roysland K, Gran JM, Ledergerber B. 2012. Causality, mediation and time: A dynamic viewpoint. *Journal of the Royal Statistical Society Series A* 175:831-861.
- Chen L, Davey SG, Harbord RM, Lewis SJ. 2008. Alcohol intake and blood pressure: A systematic review implementing a mendelian randomization approach. *PLoS Med* 5:e52.
- Daniel RM, Cousens SN, De Stavola BL, Kenward MG, Sterne JAC. 2013. Methods for dealing with time-dependent confounding. *Statistics in Medicine* 32:1584-1618.
- Greenland S, Schlesselman JJ, Criqui MH. 1987. Re: "The fallacy of employing standardized regression coefficients and correlations as measures of effect". *AJE* 125:349-350.
- Greenland S, Robins JM, Pearl J. 1999. Confounding and collapsibility in causal inference. *Statistical Science* 14:29-46.
- Greenland S, Brumback B. 2002. An overview of relations among causal modelling methods. *Int J Epidemiol* 31:1030-1037.
- Greenland S, Mansournia MA. 2015. Limitations of individual causal models, causal graphs, and ignorability assumptions, as illustrated by random confounding and design unfaithfulness. *Eur J Epidemiol*.
- Greenland SM, Malcolm; Schlesselman, James J.; Poole, Charles; Morgenstern, Hal. 1991. Standardized regression coefficients: A further critique and review of some alternatives. *Epidemiology* 2:6.
- Hafeman DM, Schwartz S. 2009. Opening the black box: A motivation for the assessment of mediation. *International Journal of Epidemiology* 38:838-845.
- Hernan MA, Hernandez-Diaz S, Werler MM, Mitchell AA. 2002. Causal knowledge as a prerequisite for confounding evaluation: An application to birth defects epidemiology. *AJE* 155:176-184.
- Hernan MA, Hernandez-Diaz S, Robins JM. 2004. A structural approach to selection bias. *Epidemiology* 15:615-625.
- Hernan MA, Cole SR. 2009. Causal diagrams and measurement bias. *AJE* 170:959-962.
- Hernan MA, Clayton D, Keiding N. 2011. The simpson's paradox unraveled. *Int J Epidemiol*.
- Hintikka J, Tolmunen T, Honkalampi K, Haatainen K, Koivumaa-Honkanen H, Tanskanen A, et al. 2005. Daily tea drinking is associated with a low level of depressive symptoms in the finnish general population. *European Journal of Epidemiology* 20:359-363.
- Lange T, Hansen JV. 2011. Direct and indirect effects in a survival context. *Epidemiology* 22:575-581.
- Mansournia MA, Hernan MA, Greenland S. 2013. Matched designs and causal diagrams. *International Journal of Epidemiology* 42:860-869.
- McCaffrey DF, Ridgeway G, Morral AR. 2004. Propensity score estimation with boosted regression for evaluating causal effects in observational studies. *Psychological Methods* 9:403-425.
- Myrstad M, Lochen ML, Graff-Iversen S, Gulsvik AK, Thelle DS, Stigum H, et al. 2014a. Increased risk of atrial fibrillation among elderly norwegian men with a history of long- term endurance sport practice. *Scand J Med Sci Spor* 24:E238-E244.

- Pearl J. 2000. Causality: Models, reasoning, and inference. Cambridge:Cambridge University Press.
- Pearl J. 2012. The causal mediation formula-a guide to the assessment of pathways and mechanisms. *Prev Sci* 13:426-436.
- Robins JM, Greenland S. 1992. Identifiability and exchangeability for direct and indirect effects. *Epidemiology* 3:143-155.
- Robins JM. 2001. Data, design, and background knowledge in etiologic inference. *Epidemiology* 12:313-320.
- Robinson LD, Jewell NP. 1991. Some surprising results about covariate adjustment in logistic-regression models. *Int Stat Rev* 59:227-240.
- Rothman KJ, Greenland S, Lash TL. 2008. Modern epidemiology. Philadelphia:Lippincott Williams & Wilkins.
- Shahar E. 2009. Causal diagrams for encoding and evaluation of information bias. *Journal of evaluation in clinical practice* 15:436-440.
- Shahar E, Shahar DJ. 2012. Causal diagrams and the logic of matched case-control studies. *Clinical epidemiology* 4:137-144.
- Sheehan NA, Didelez V, Burton PR, Tobin MD. 2008. Mendelian randomisation and causal inference in observational epidemiology. *PLoS Med* 5:e177.
- Textor J, Hardt J, Knuppel S. 2011. Dagitty a graphical tool for analyzing causal diagrams. *Epidemiology* 22:745-745.
- VanderWeele TJ, Robins JM. 2007. Directed acyclic graphs, sufficient causes, and the properties of conditioning on a common effect. *AJE* 166:1096-1104.
- VanderWeele TJ, Hernan MA, Robins JM. 2008. Causal directed acyclic graphs and the direction of unmeasured confounding bias. *Epidemiology* 19:720-728.
- VanderWeele TJ. 2009. Mediation and mechanism. *Eur J Epidemiol* 24:217-224.
- VanderWeele TJ, Arah OA. 2011. Bias formulas for sensitivity analysis of unmeasured confounding for general outcomes, treatments, and confounders. *Epidemiology* 22:42-52.
- VanderWeele TJ, Hernan MA. 2012. Results on differential and dependent measurement error of the exposure and the outcome using signed directed acyclic graphs. *AJE* 175:1303-1310.
- VanderWeele TJ. 2014. A unification of mediation and interaction: A 4-way decomposition. *Epidemiology* 25:749-761.
- Veieroed M, Lydersen S, Laake P. 2012. Medical statistics in clinical and epidemiological research. Oslo:Gyldendal Akademisk.
- Westreich D, Greenland S. 2013. The table 2 fallacy: Presenting and interpreting confounder and modifier coefficients. *AJE* 177:292-298.
- Xing C, Xing GA. 2010. Adjusting for covariates in logistic regression models. *Genet Epidemiol* 34:937-937.

¡Gracias!